

Shorting Private Equity: A Structural and Cyclical Call

Barry Zhang, Krish Pillai, Bingran Li, Urmil Shah, Nandini Goel

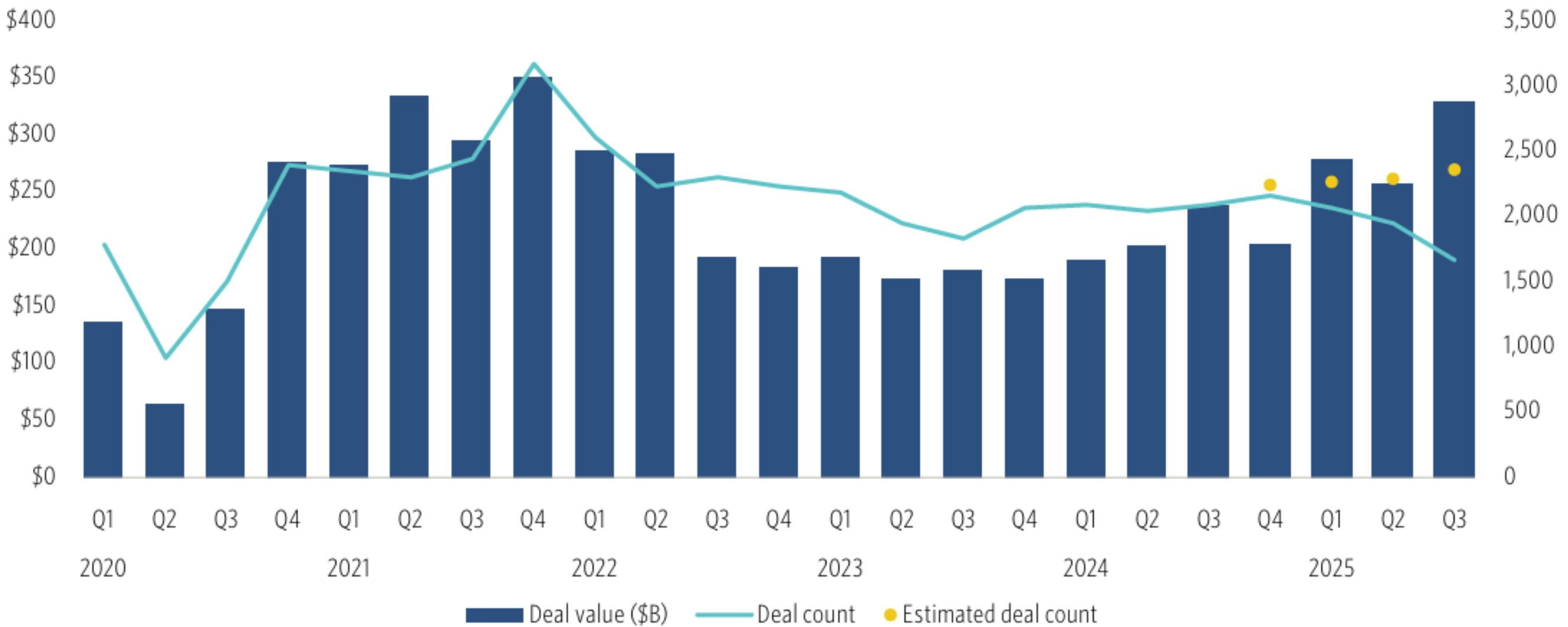
Section 2 | Thursday 3:30 pm

Investment Thesis

Despite rising interest rates and tighter credit, private equity industry deal multiples and valuations remain at record highs. Concentrated portfolios face valuation compression and exit challenges. Excessive leverage and inflated entry prices heighten the risk of significant markdowns, making “PSP” a strategic short in a deteriorating private equity cycle.

US Private Equity Deal Activity

Deal activity increased in Q3 2025, marking both a QoQ uptick and YoY increase.

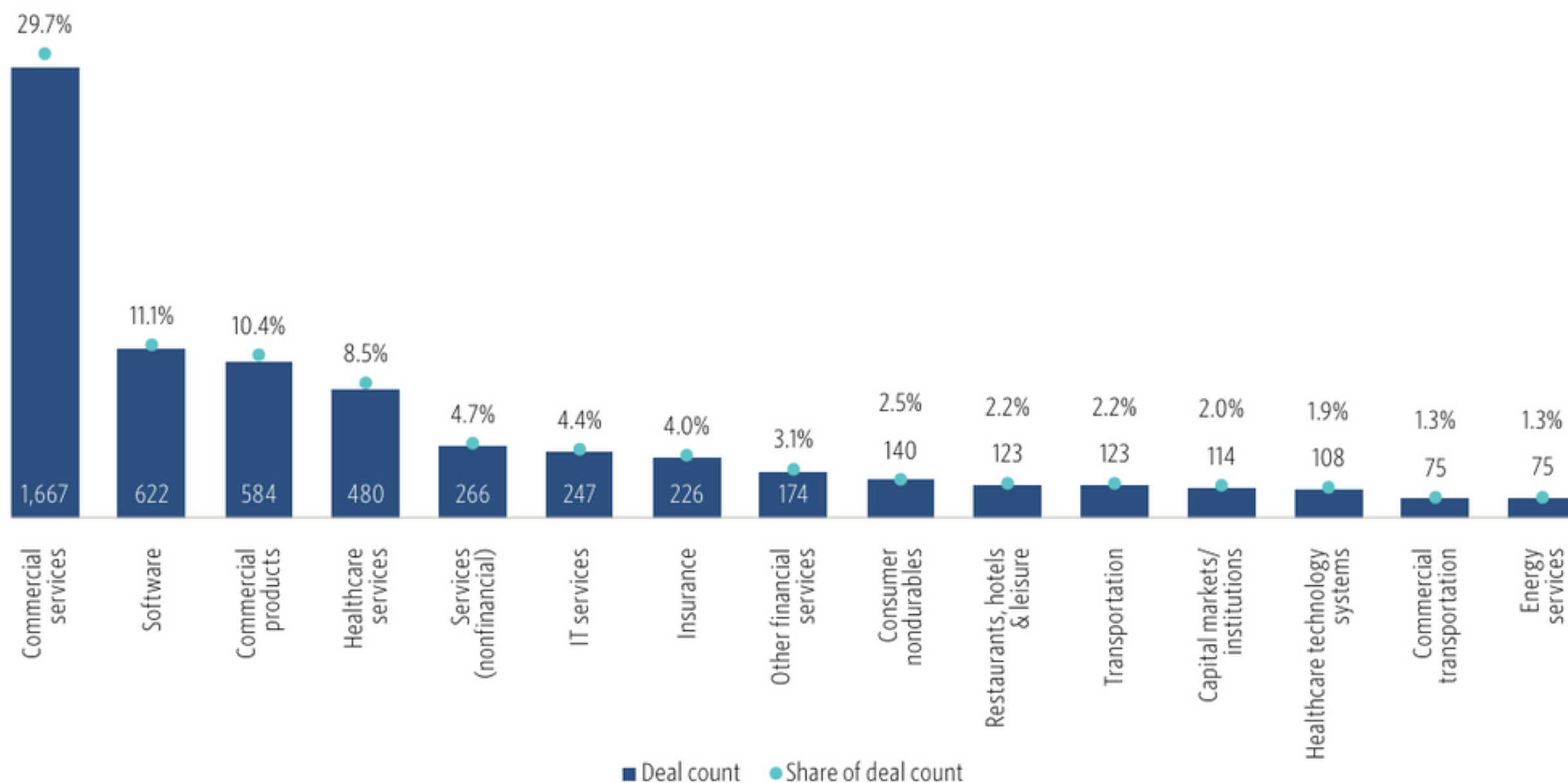


Source: Pitchbook; <https://pitchbook.com/news/reports/q3-2025-us-pe-breakdown>

Assumption

- U.S. private equity deal value surged from around \$200 billion in Q1 2024 to nearly \$330 billion by Q3 2025, supported by mega buyouts and renewed public-to-private transactions.
- Despite this rebound, valuations remain stretched as leverage costs rise and liquidity thins.
- The volume expansion doesn't offset structural vulnerabilities, persistent rate pressure, limited exits, and aging portfolios, sustaining PSP's short-side appeal.

Top 15 industry groups by PE deal count in 2025



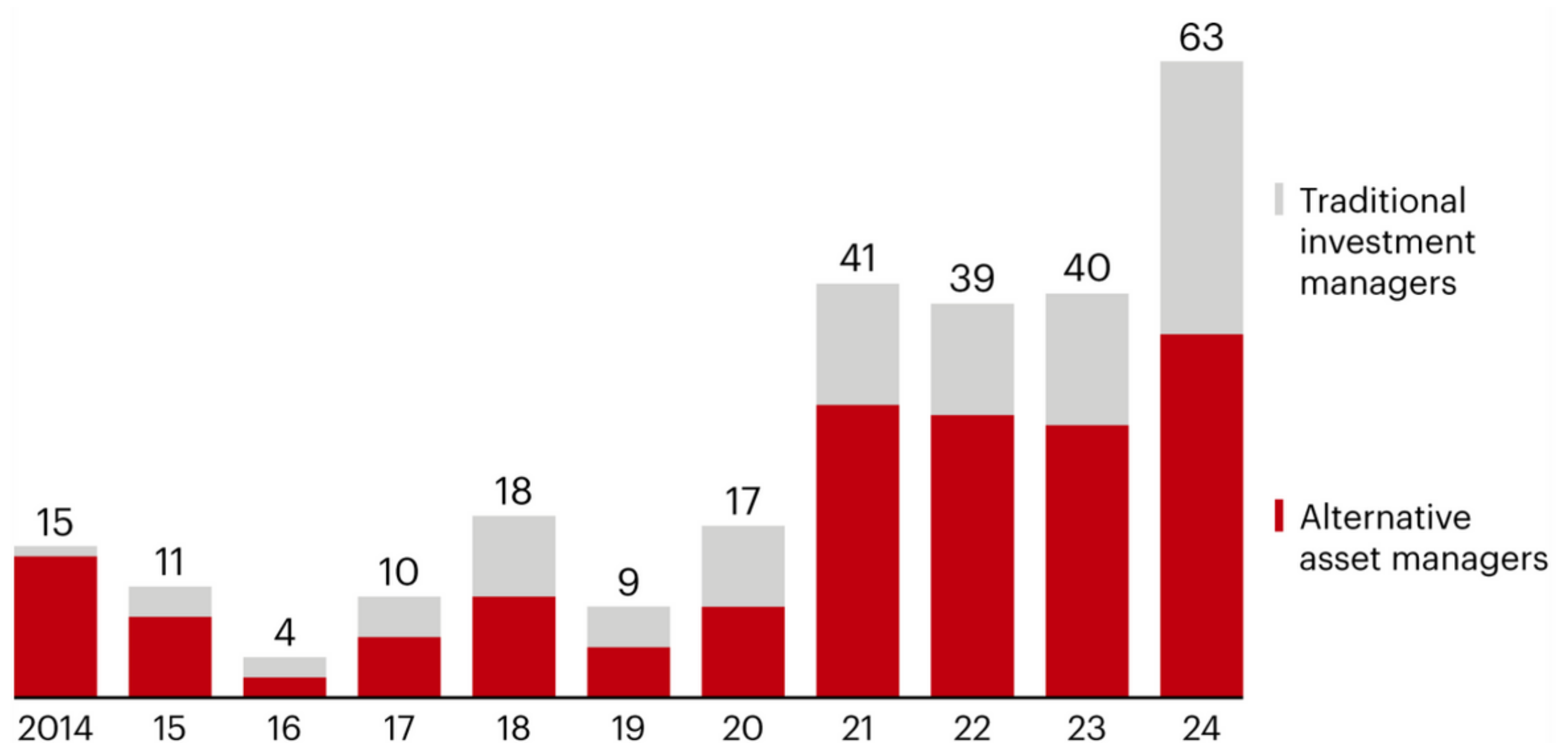
Source: Pitchbook; <https://pitchbook.com/news/reports/q3-2025-us-pe-breakdown>

Assumption

- U.S. private equity deal count rose modestly in 2025, reflecting renewed activity in large add-on acquisitions and take-private transactions.
- However, despite higher transaction volumes, valuations remain inflated and leverage costs elevated.
- This surge stems from funds deploying excess dry powder, not from healthier fundamentals.
- Aging portfolios, limited exits, and tightening credit still justify maintaining a short on PSP.

M&A in Private Equity has Taken Off

Most firms used to lack the balance sheet to fund acquisitions.



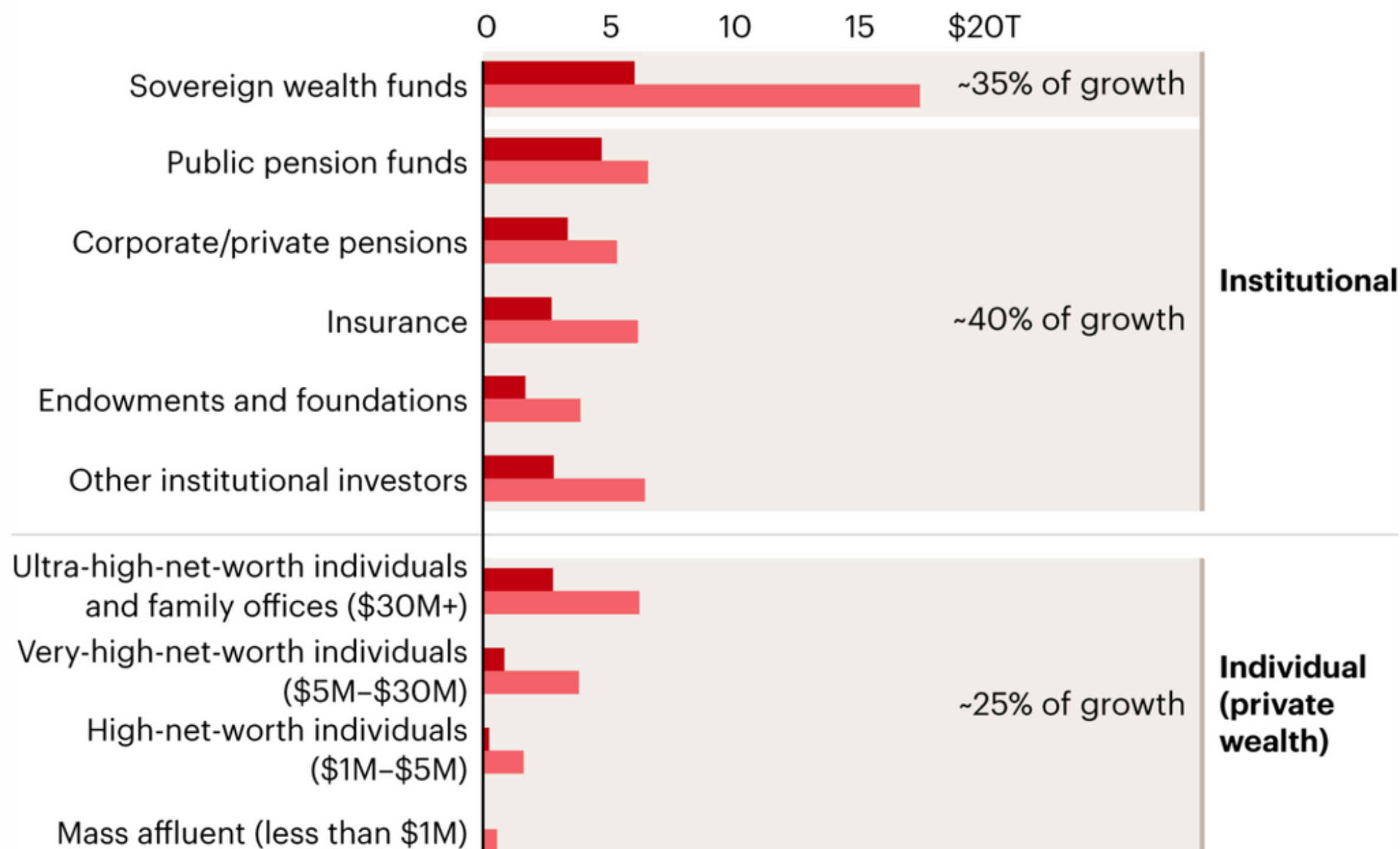
Source: Bain Strategic M&A in Alternatives Database; <https://www.bain.com/insights/strategic-imperative-in-private-equity-global-private-equity-report-2025/>

Assumption

- M&A activity in private equity accelerated sharply in 2025, with deal value up over 33% year-over-year and PE firms driving nearly 60% of U.S. transactions above \$100 million.
- Fueled by record dry powder, rate cuts, and easing regulation, sponsors executed mega-deals across tech, healthcare, and energy.
- Yet this momentum masks valuation inflation, leverage risks, and exit bottlenecks, conditions that continue to validate a short thesis on PSP.

Future Growth in Assets

■ 2023 ■ 2033



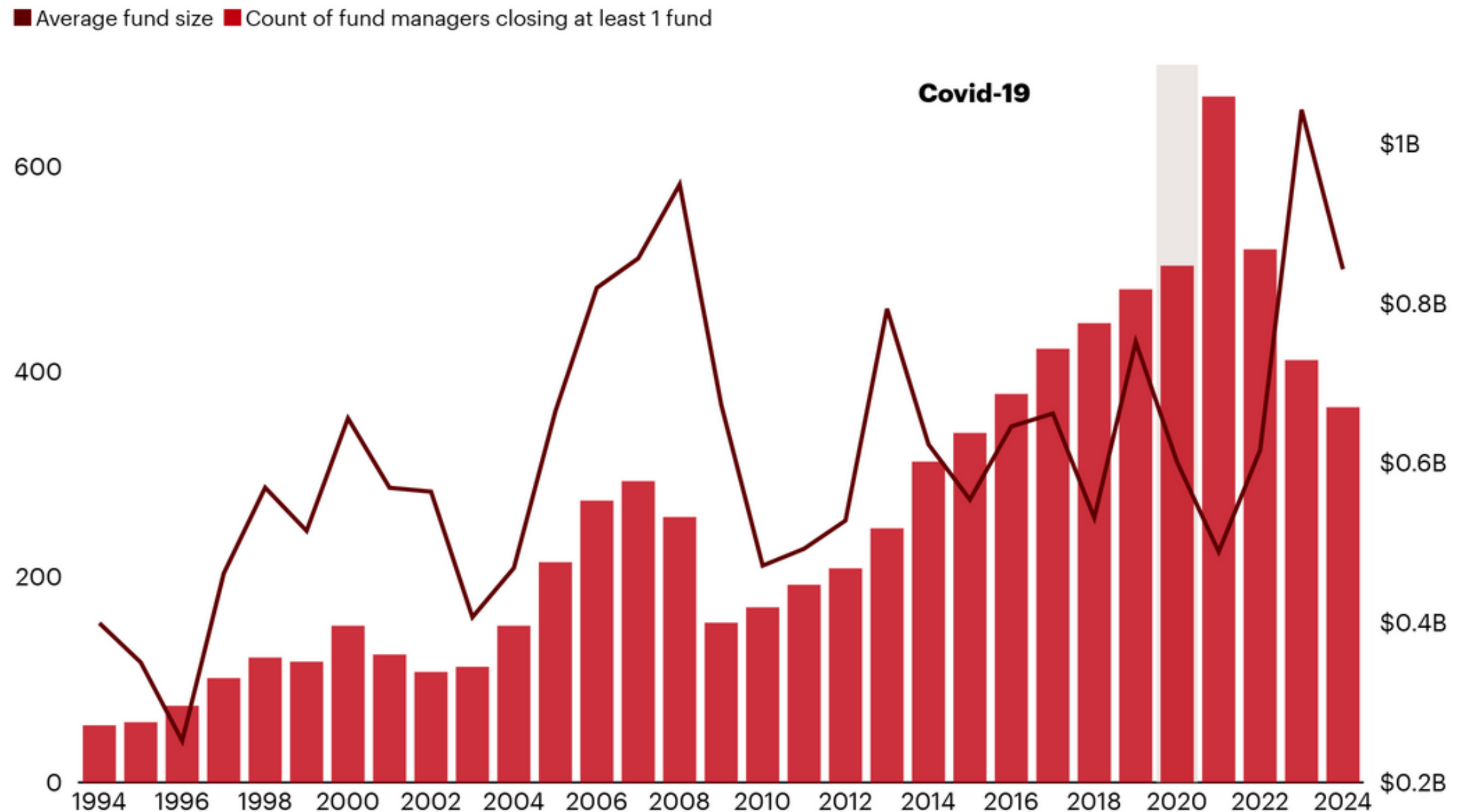
Source: <https://www.bain.com/insights/strategic-imperative-in-private-equity-global-private-equity-report-2025/>

Assumption

- U.S. private equity assets under management have shown robust growth, with PSP Investments reporting net AUM nearing \$300 billion in fiscal 2025, a 13.2% increase year-over-year driven largely by \$33.5 billion in net portfolio income.
- Despite economic uncertainty and inflationary pressures, PSP's diversified portfolio, including infrastructure, private equity, credit, and real estate, supports its growth outlook.
- However, challenges like valuation pressures and liquidity constraints persist. Even with asset growth, risks of inflated entry valuations and exit bottlenecks sustain a cautious view on PSP's future performance, supporting a short thesis in the current market environment.

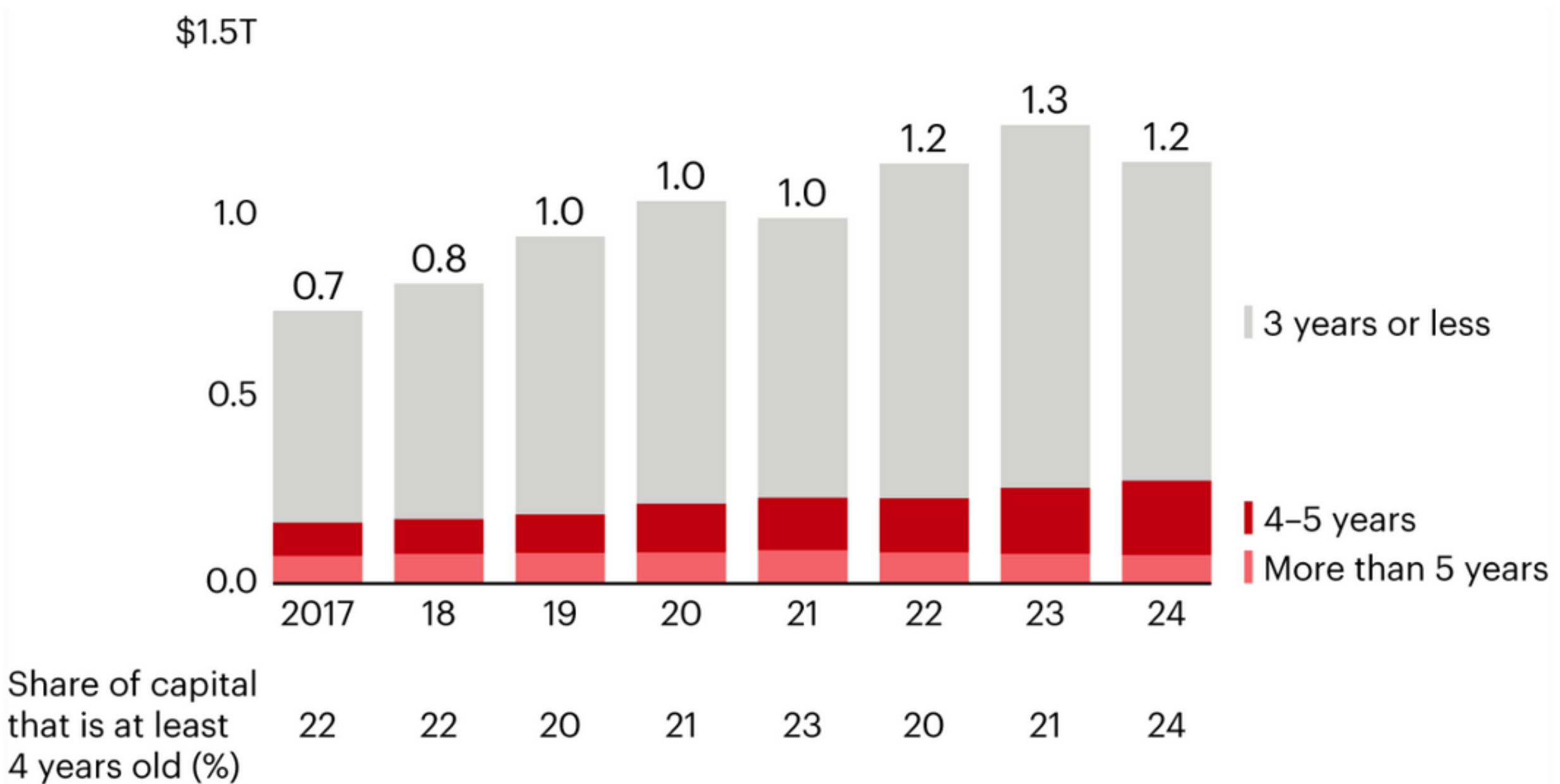
Average Fund Size Decreasing Since 2021.

Economic Downturns Typically Reduce the Number of Funds Seeking Capital.



Source: <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report?>

Global Buyout Dry Powder



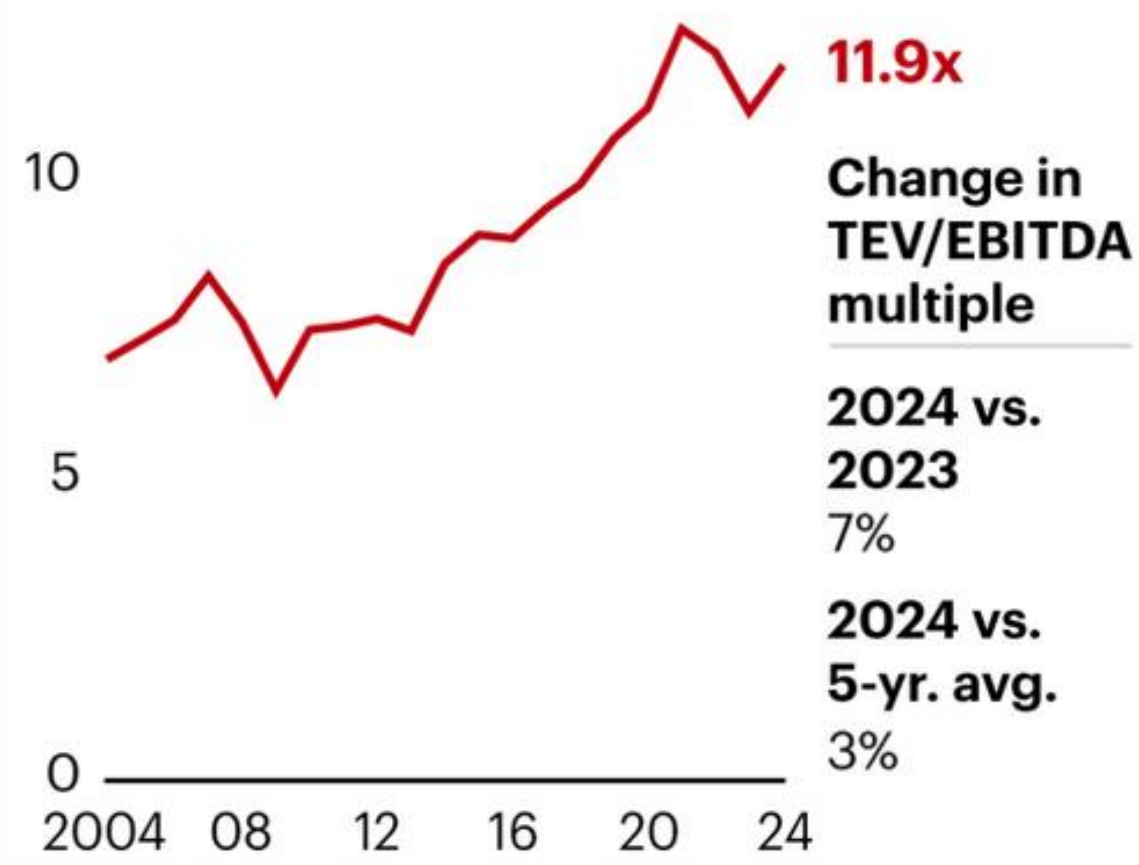
Source: <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report?>

Deal Multiples Remain at Record Levels

North America

Median total enterprise value (TEV)/EBITDA multiple

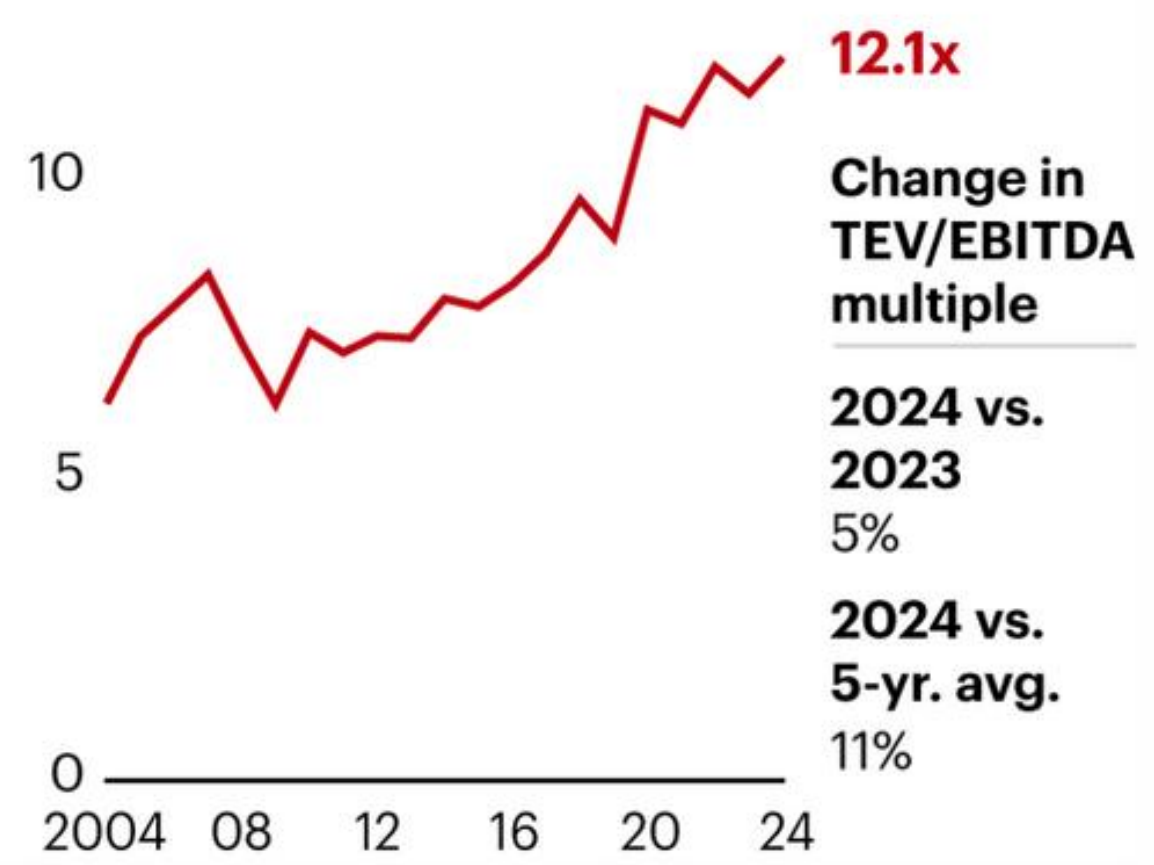
15x



Western Europe

Median TEV/EBITDA multiple

15x



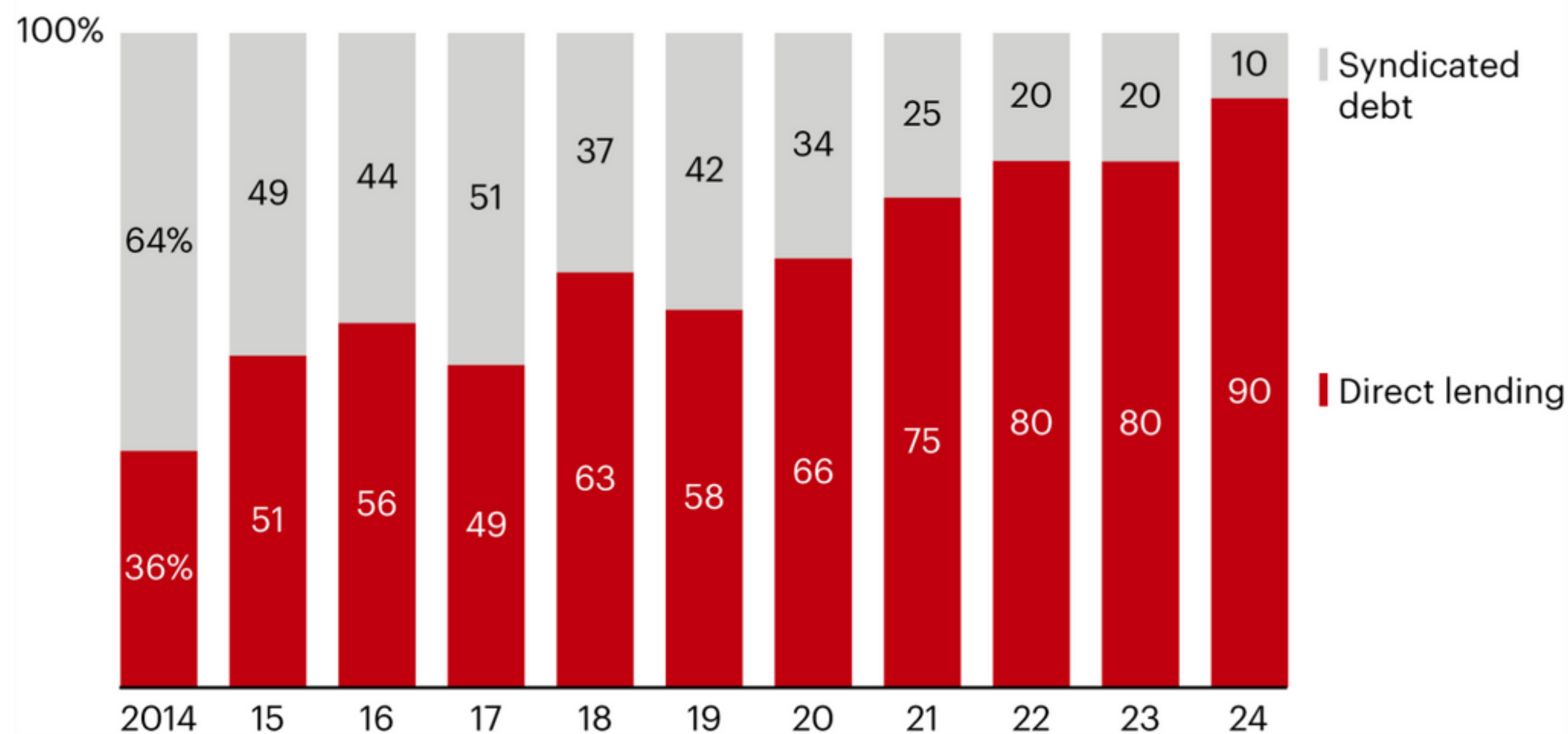
Source: <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report?>

Assumption

- In 2025, private equity deal multiples have remained at historically high levels, with median EBITDA purchase multiples hovering around 12x to 15x, reflecting elevated valuations amid intense competition for quality assets.
- Despite rising interest rates and credit costs, sponsors continue to pay premiums fueled by abundant dry powder and strong fundraising.
- This heightens the risk of multiple compression and valuation corrections for PSP's portfolio companies, reinforcing downside risks and validating a short position given the vulnerability of inflated deal prices in a tightening macroeconomic environment.

Direct Lending Continues its Growth

Share of US middle-market leveraged buyout loan issuance, by debt type

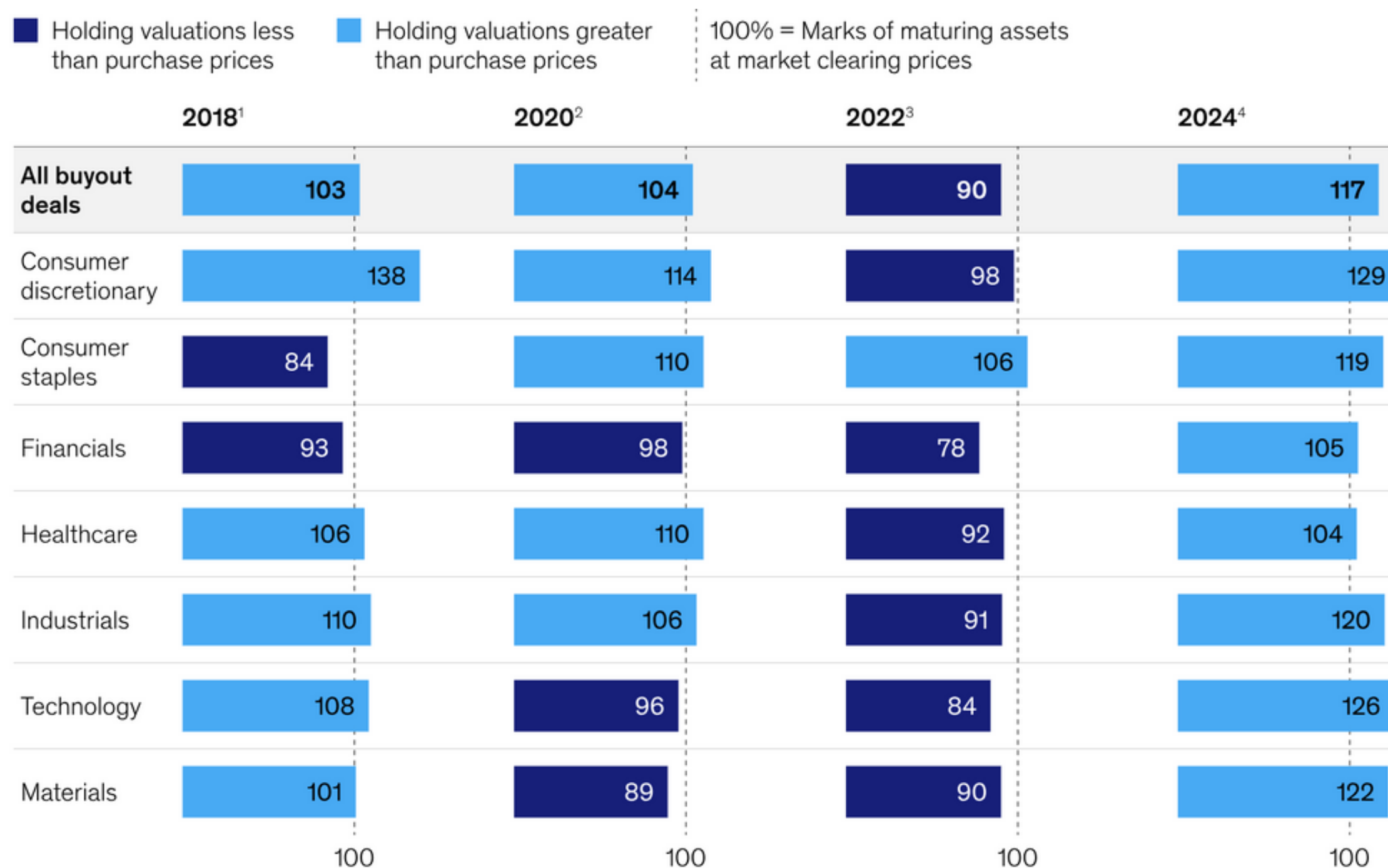


Source: <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report?>

Assumption

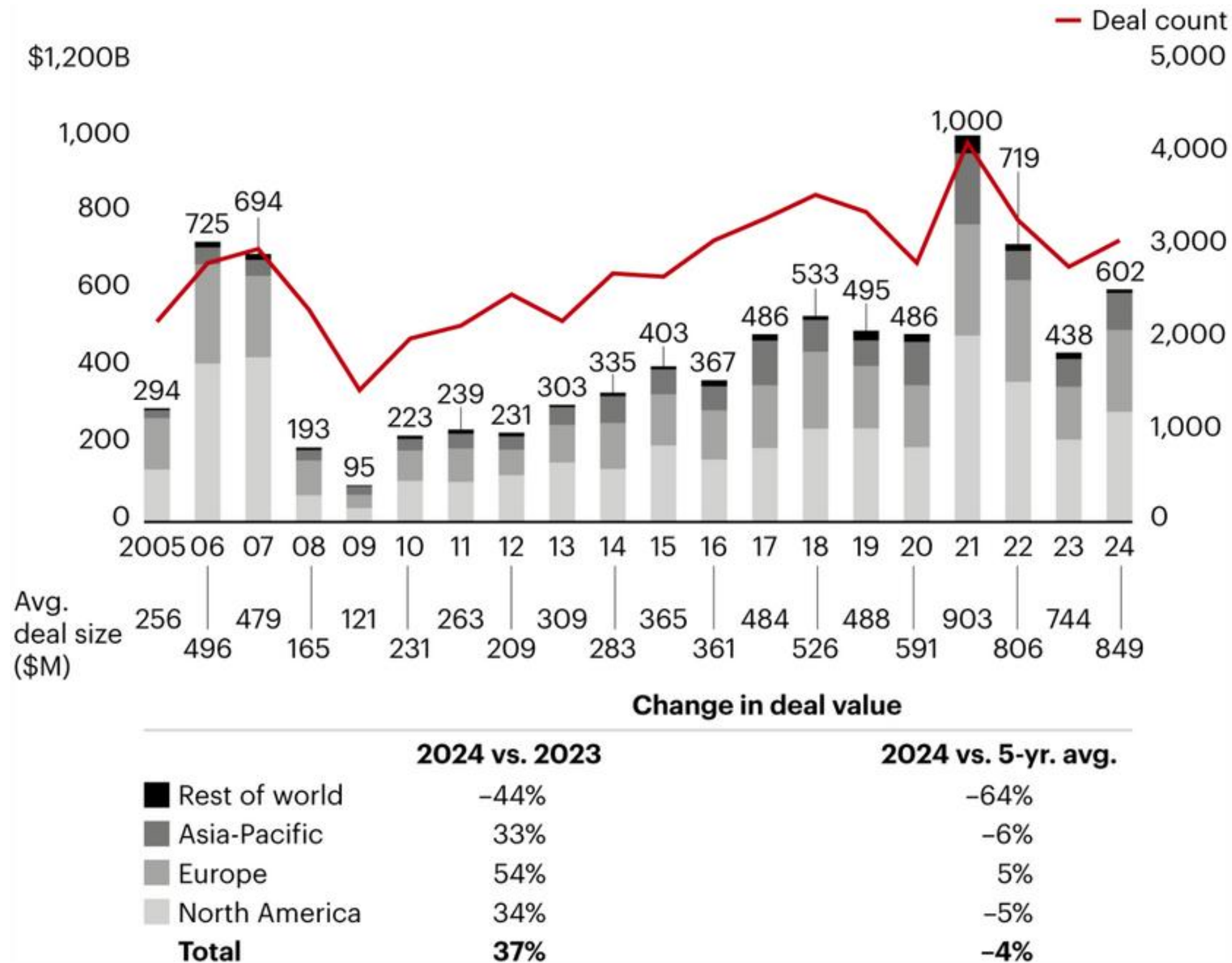
- Direct lending has shown strong growth in 2025, with assets under management increasing as private equity firms and institutional investors target higher-yield credit opportunities amid tighter bank lending standards.
- Favorable yields and flexible terms attract borrowers, supporting continued expansion despite macroeconomic headwinds.
- However, rising default risks and interest rate volatility pose challenges. For PSP, exposure to growing direct lending may enhance income but also increase credit risk, underscoring caution in the short thesis due to potential impairments and valuation pressures in this sector.

Near-maturity assets are increasingly held at valuations higher than the prevailing market price.



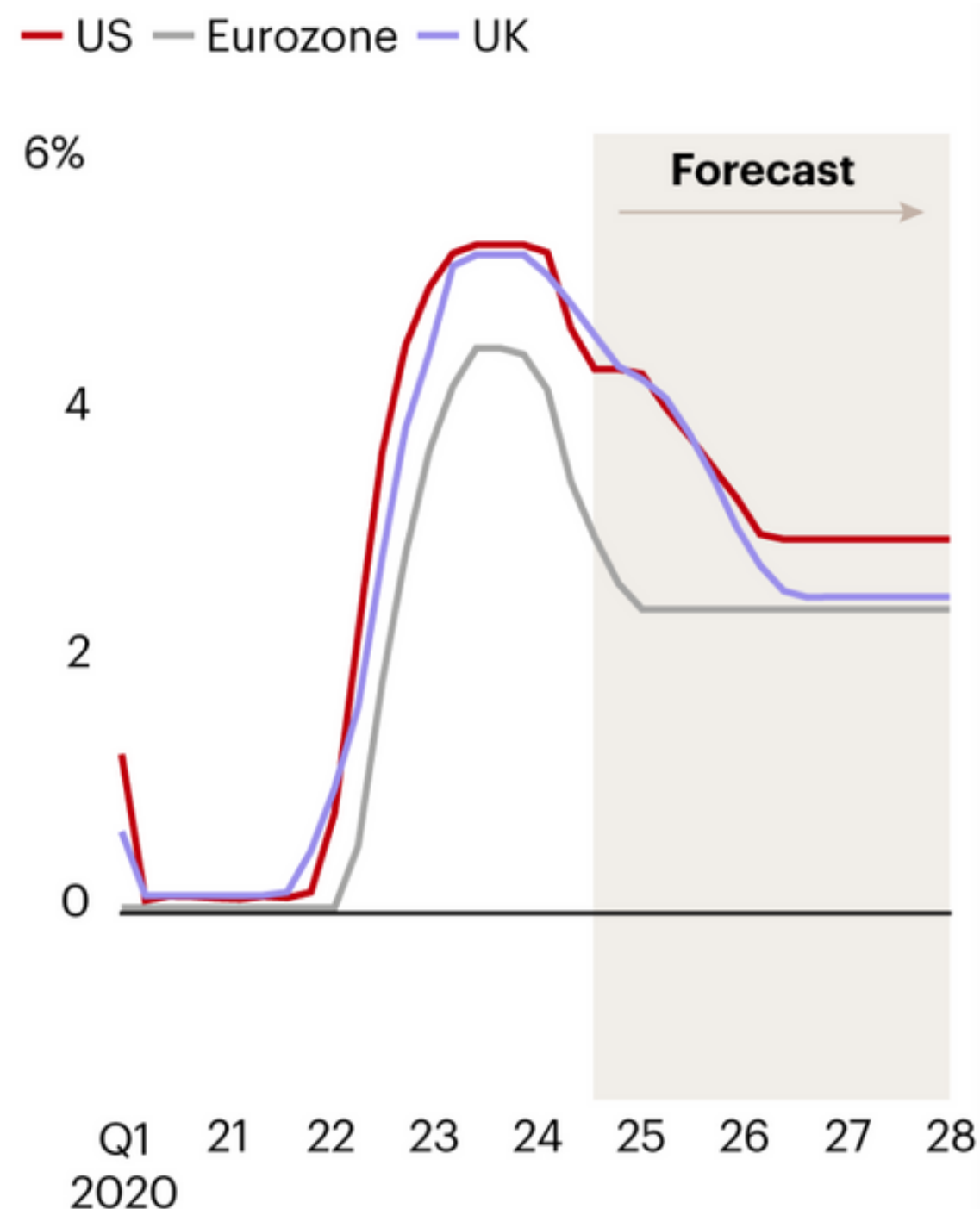
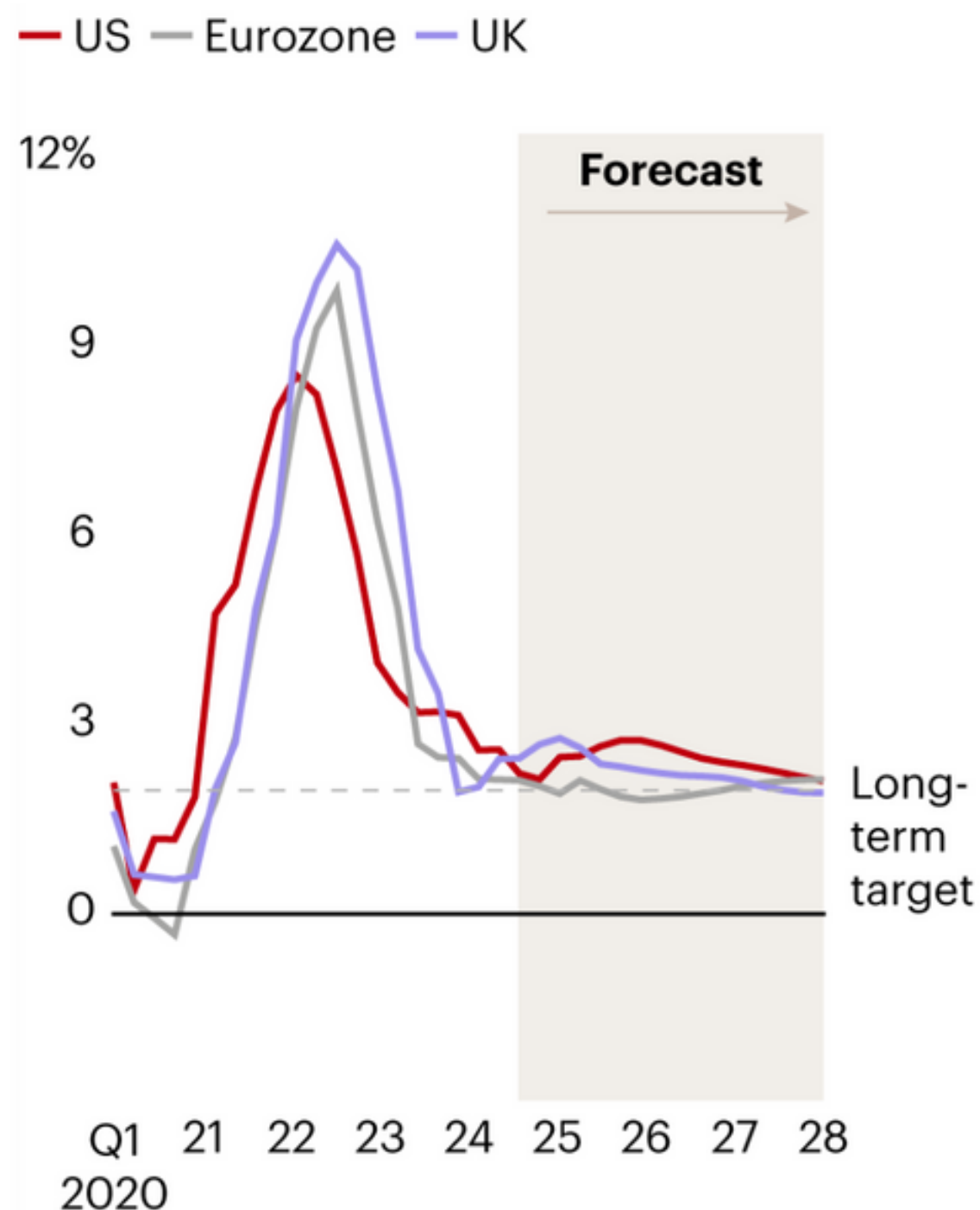
Source: <https://www.mckinsey.com/industries/private-capital/our-insights/global-private-markets-report?>

Global buyout deal value, by region



Source: Dealogic; <https://www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025/>

Interest Rates Expected to Continue Cooling



Source: Moody's; US Bureau of Labor Statistics; European Commission; UK Office for National Statistics; US Board of Governors of the Federal Reserve System; European Central Bank; Bank of England; <https://www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025/>

Interest Rates & Credit Market Evolution

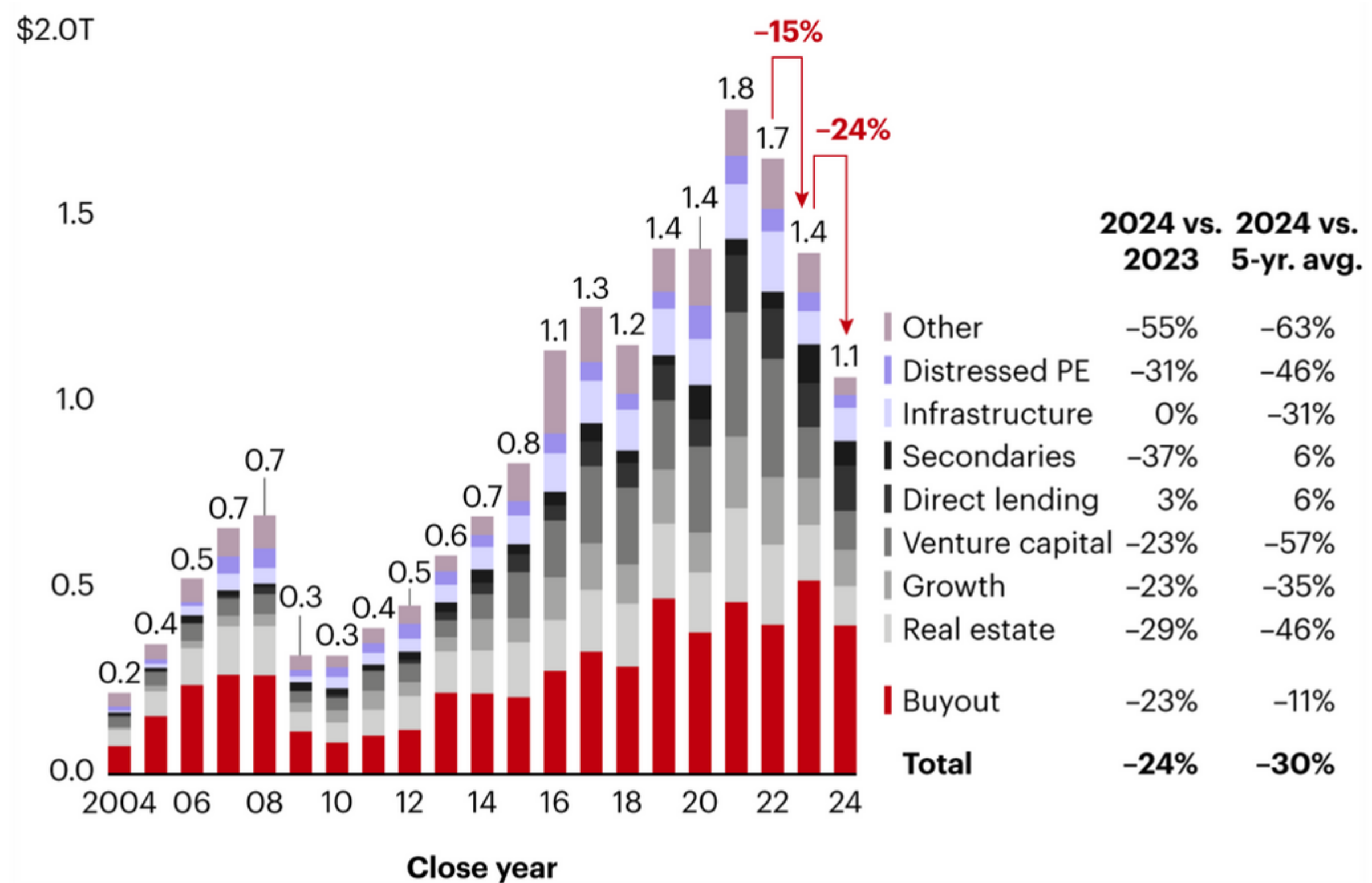
- The Federal Reserve's rate cuts since mid-2024 have fueled optimism, but easy credit can unwind quickly if inflation or defaults rise again.
- Falling rates may already be priced in, leaving limited upside for listed private-equity stocks like those in PSP.
- Weaker lending standards and slower deal exits could pressure valuations if markets turn risk-off.
- A reversal toward higher rates or wider credit spreads would raise financing costs and weigh on PSP's earnings and share price.

Interest Rates & Credit Market Evolution

Year	Fed Funds Rate (%)	U.S. High-Yield OAS (%)	Credit Market Trend	PE Deal Activity
2022	4.25	5.8	Tight – Sharp rise in cost of capital	–40%
2023	5.25	4.5	Restrictive	–15%
2024	4.75	3.5	Temporary easing	37%
2025 (E)	4.25	3	Fragile	–15 % to –20 % (expected slowdown)

Source: <https://www.macrotrends.net/2015/fed-funds-rate-historical-chart>

Global Private Fundraising Continues to Decline

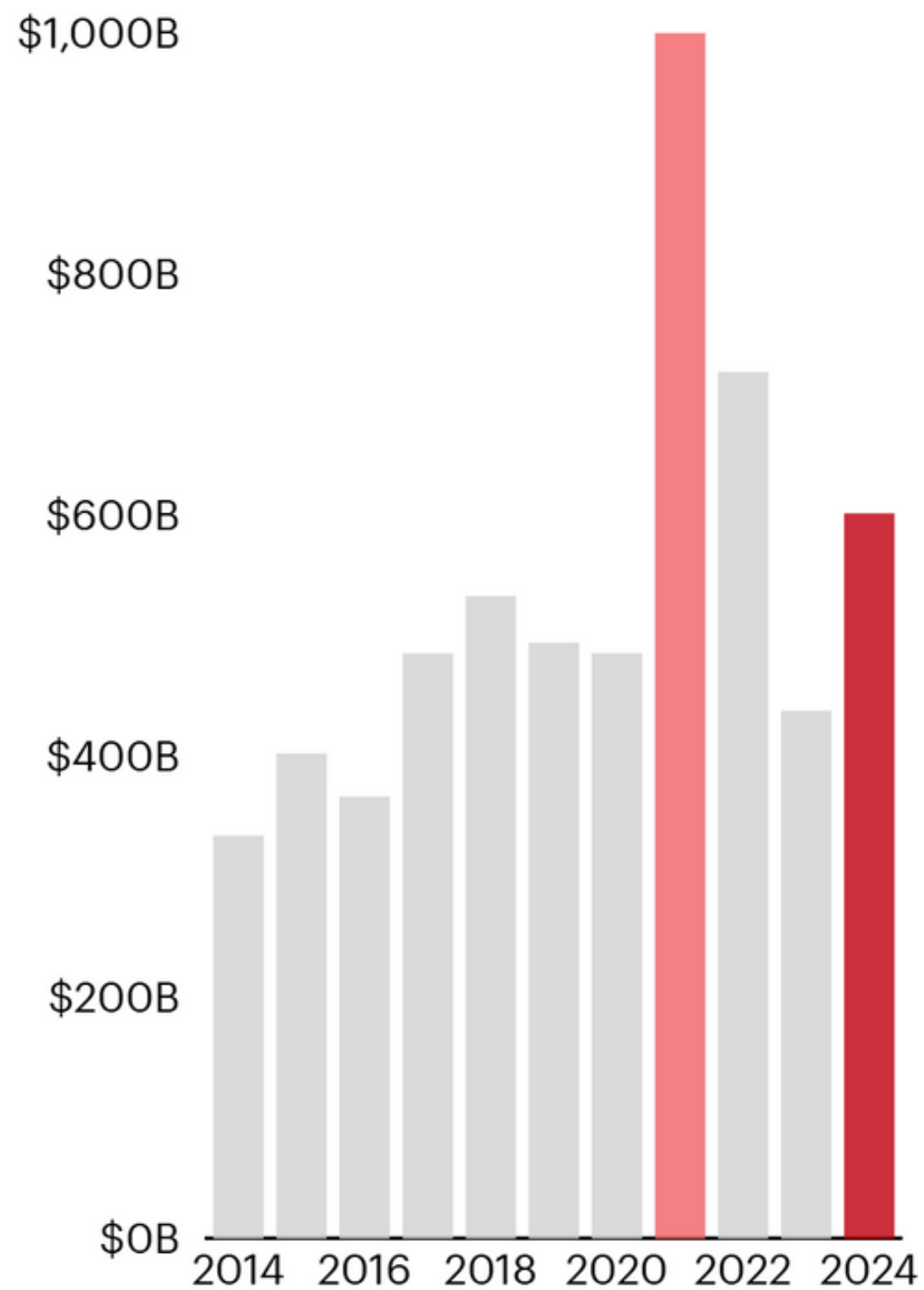


Source: Preqin; <https://www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025/>

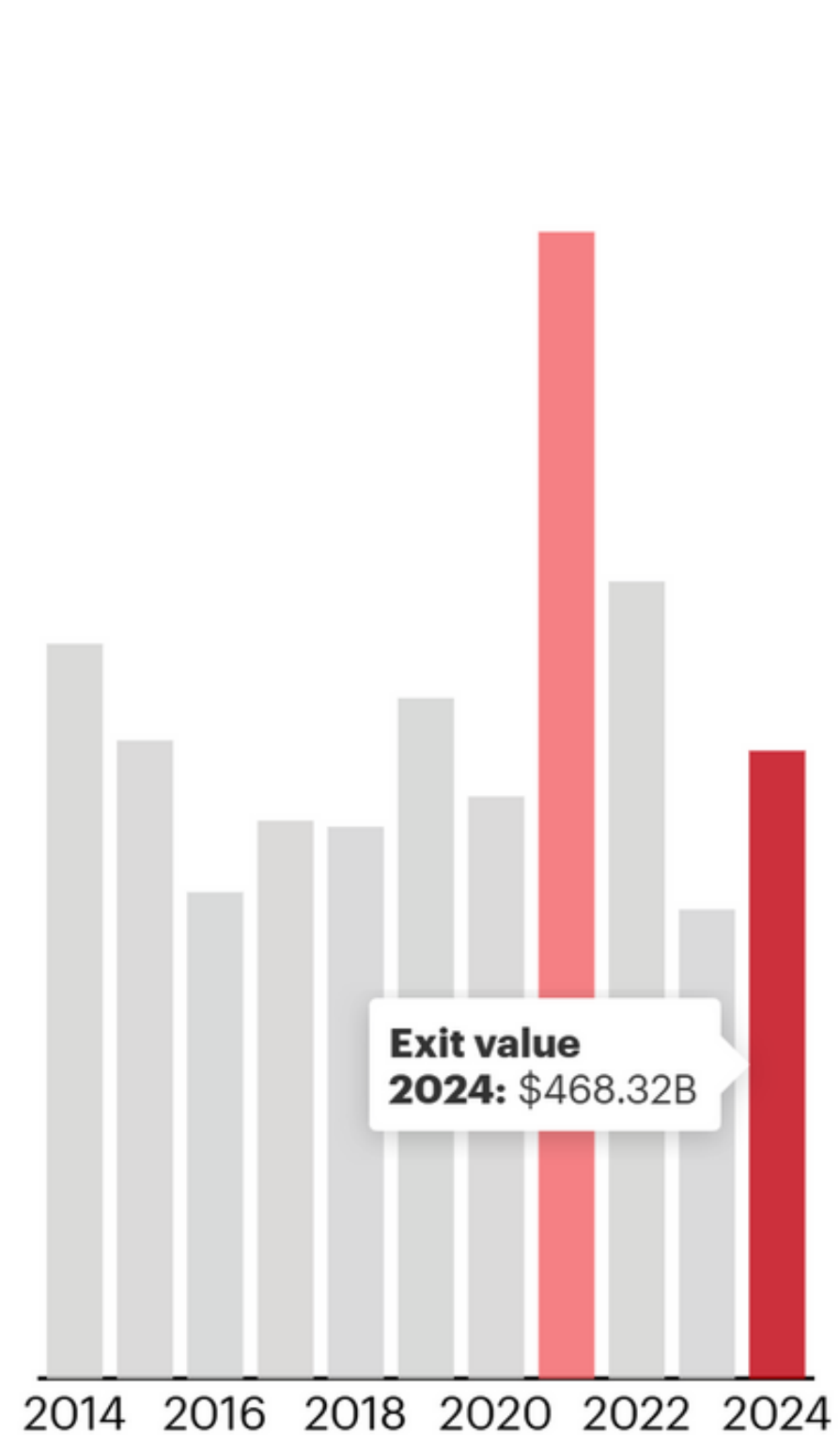
Fundraising Trends

- Fundraising activity in private equity has fallen sharply since the 2021 peak, when over \$1.15 trillion was raised across 6,800 funds.
- The 2022–2024 period saw a 40%+ drop in capital raised as higher borrowing costs and slower exits kept investors cautious.
- However, 2025 shows early signs of stabilization, fundraising of around \$425 billion across 1,000 funds suggests that confidence is returning, particularly among large global sponsors.
- This trend signals a potential re-acceleration of capital flows and deal activity, which supports earnings growth for listed private-equity managers held within the PSP ETF.

Investment Value

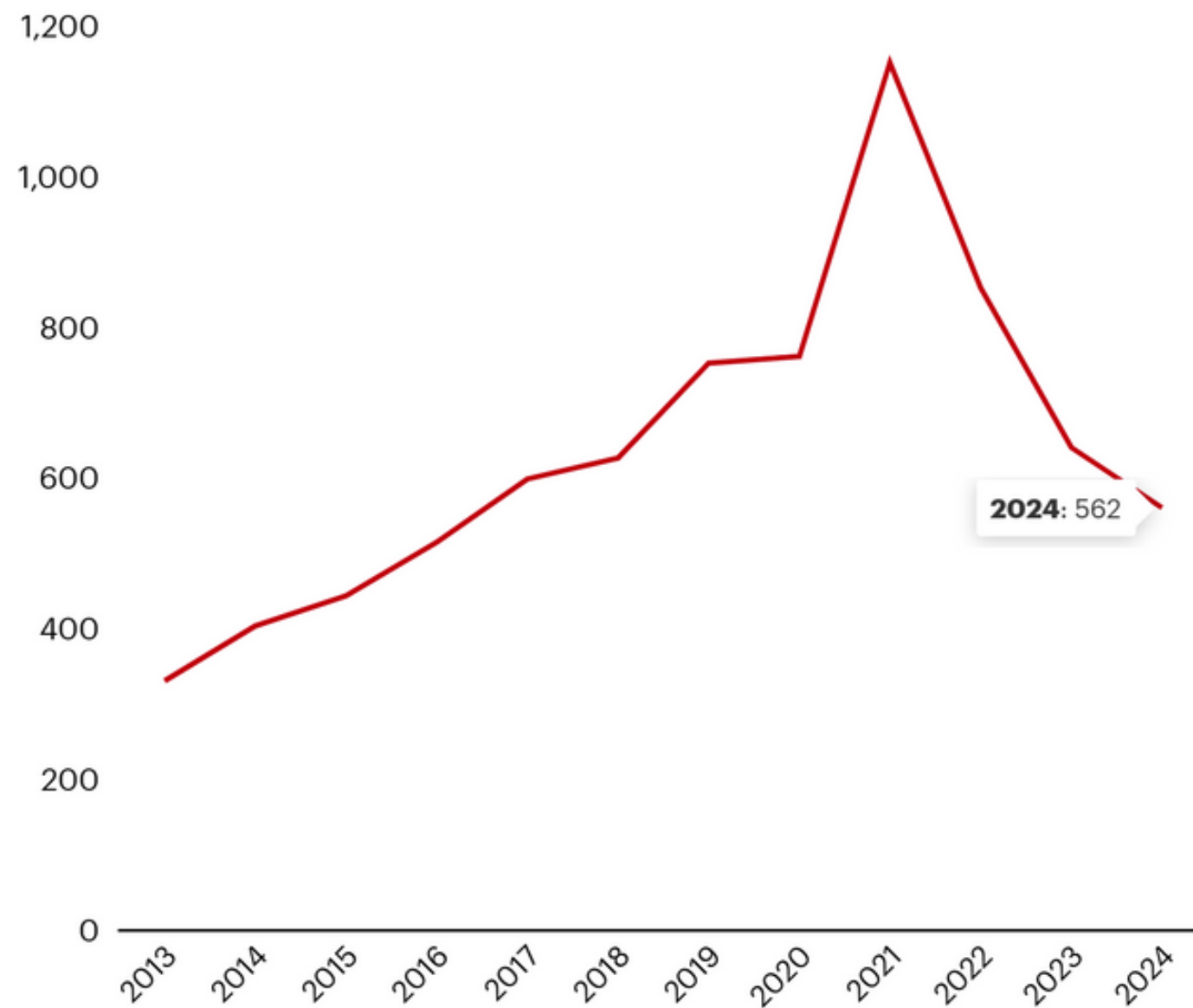


Exit Value

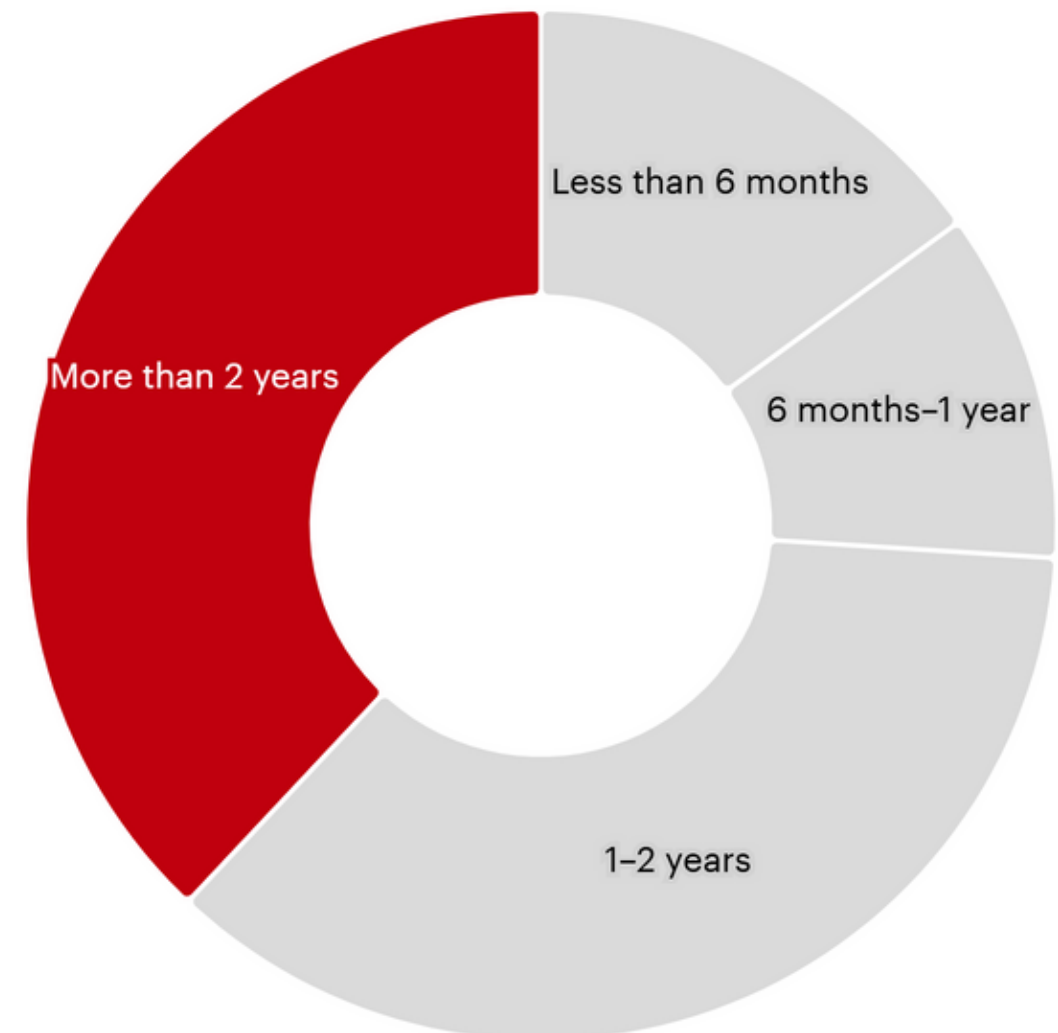


Source: Dealogic; Preqin; <https://www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025/>

Count of Funds Closed



Time to Close

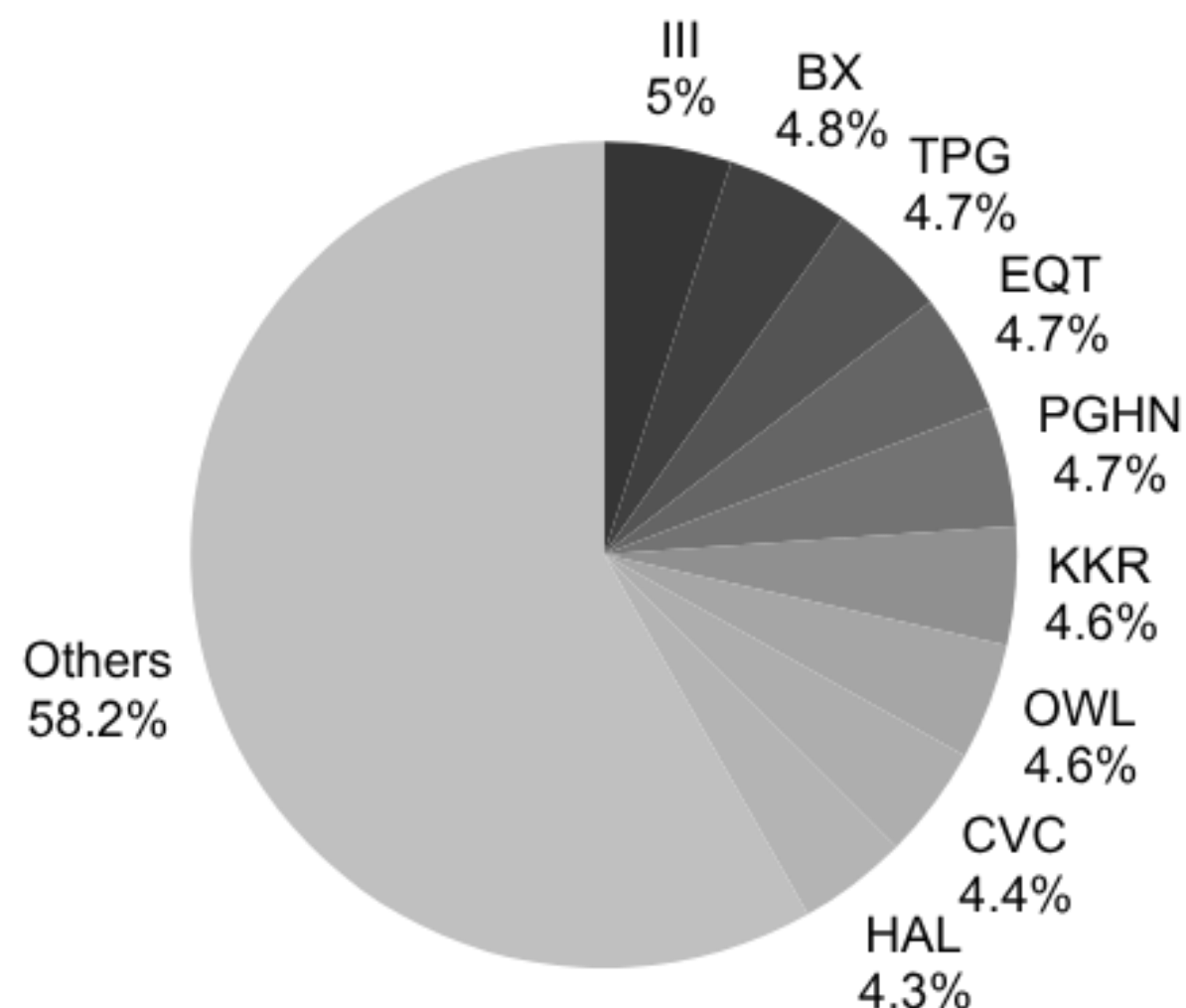


Source: Dealogic; Preqin; <https://www.bain.com/insights/outlook-is-a-recovery-starting-to-take-shape-global-private-equity-report-2025/>

PSP Portfolio Composition

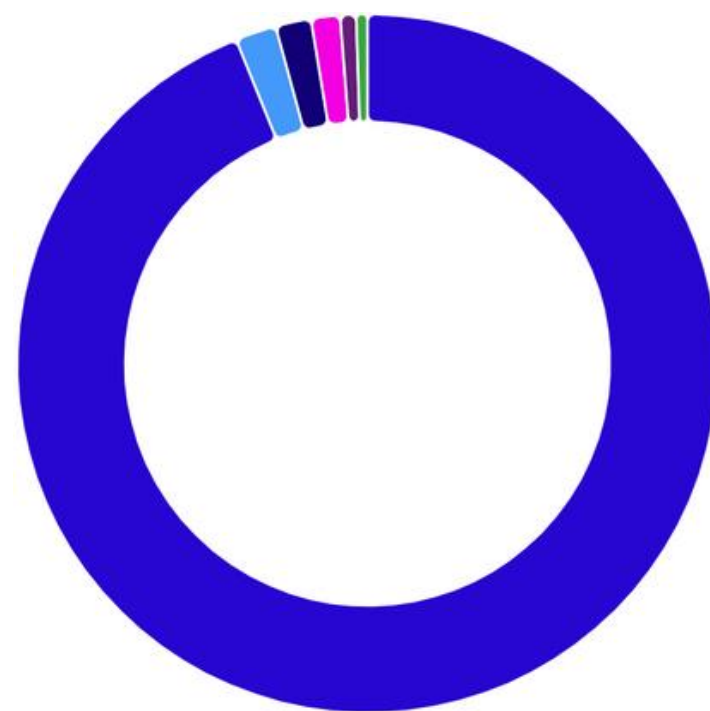
Top 10 Holdings:

Ticker	Company	Wt
III	3i Group PLC	4.99%
BX	Blackstone Inc	4.82%
TPG	TPG Inc	4.71%
EQT	EQT	4.70%
PGHN	Partners Group Holding	4.69%
KKR	KKR & Co Inc	4.60%
OWL	Blue Owl Capital	4.57%
CVC	CVC Capital Partners	4.39%
HAL	HAL Trust	4.31%
SFNX	Sofina	4.29%

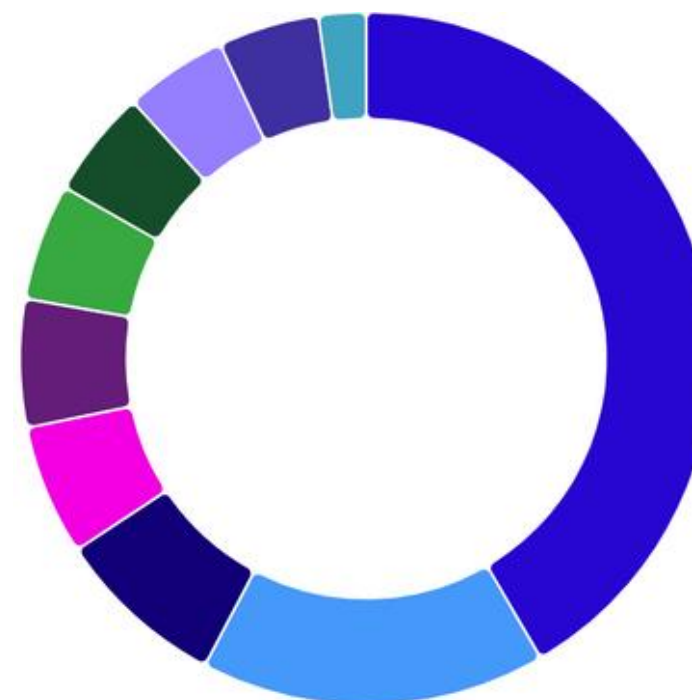


Source: <https://www.invesco.com/us/en/financial-products/etfs/invesco-global-listed-private-equity-etf.html>

PSP Portfolio Sector & Geographical Composition



Financials	93.89%
Industrials	1.89%
Consumer discretionary	1.63%
Consumer staples	1.35%
Communication services	0.72%
Information technology	0.52%

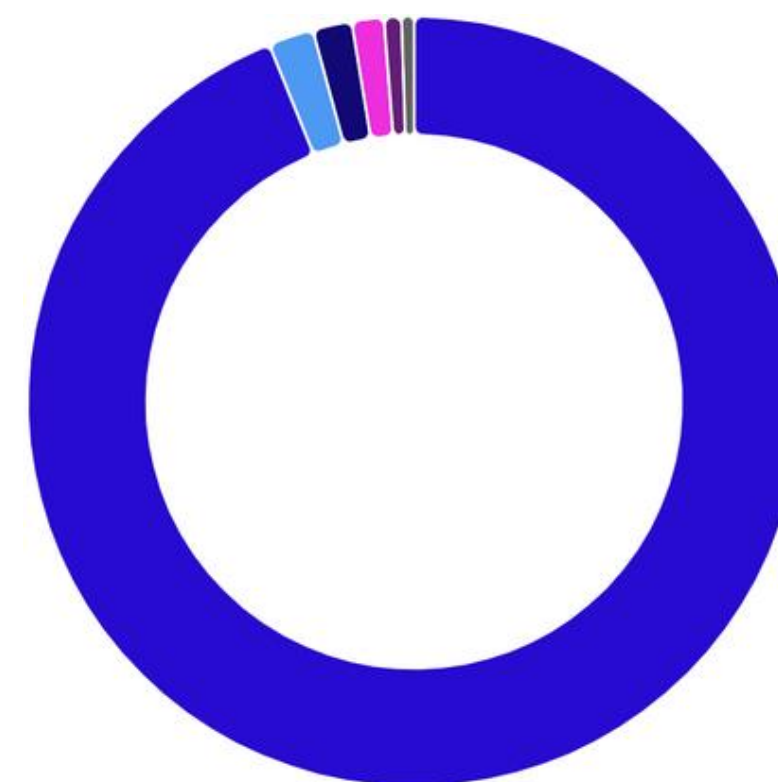


United States	38.05%
United Kingdom	14.75%
Sweden	7.31%
Belgium	5.58%
Switzerland	5.46%
France	4.98%
Canada	4.62%
Luxembourg	4.39%
Netherlands	4.31%
Guernsey	2.03%

Source: <https://www.invesco.com/us/en/financial-products/etfs/invesco-global-listed-private-equity-etf.html>

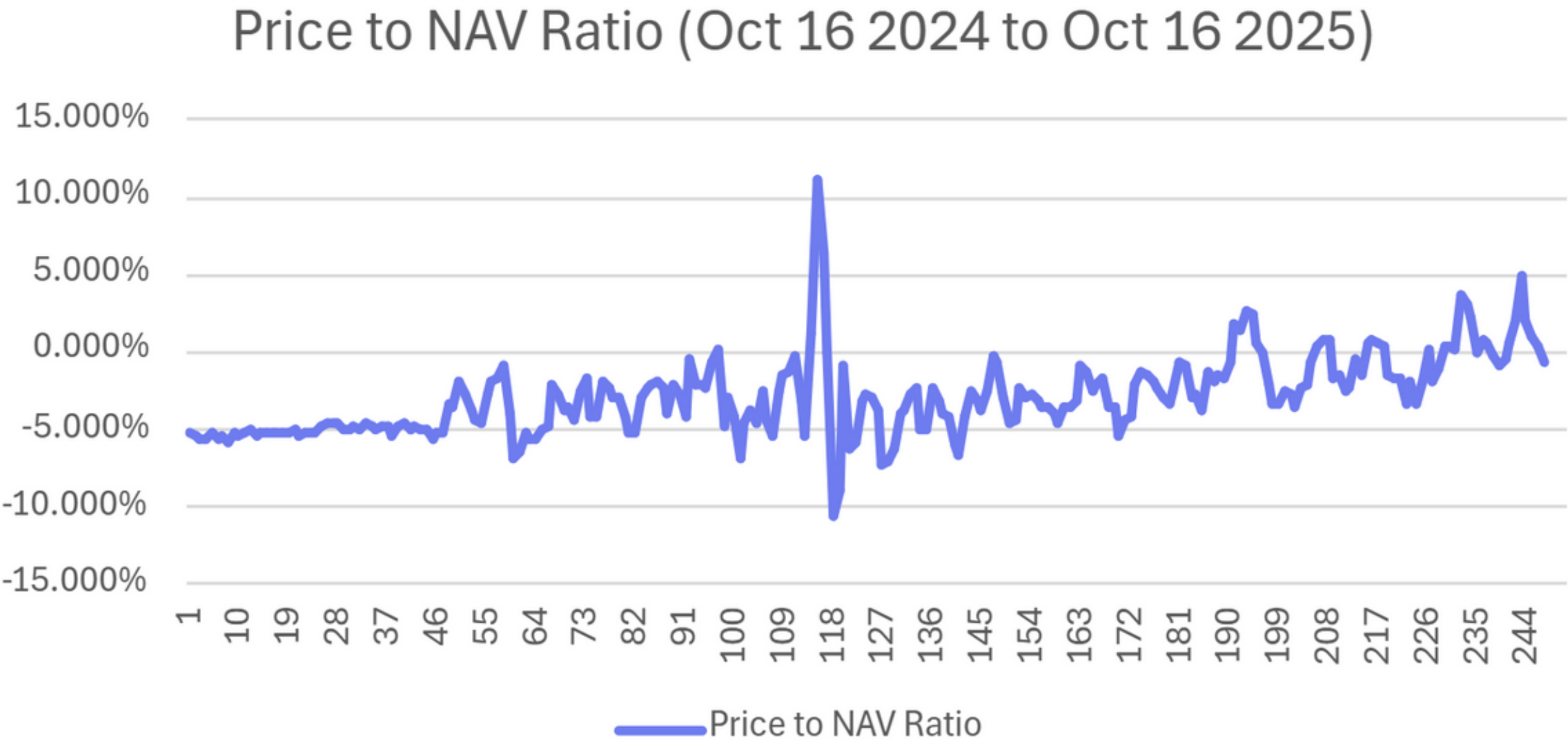
Concentration Risk Analysis (Top Holdings Focus)

- PSP's portfolio is heavily concentrated in the Financials sector (93.9%), leaving minimal exposure to other industries. This limits diversification benefits and increases concentration risk.
- Top 10 holdings account for 41.78% of PSP's total portfolio, reflecting moderate to high concentration.
- Portfolio is primarily concentrated in developed European and North American markets, which leads to reduced global diversification.



Financials	93.89%
Industrials	1.89%
Consumer discretionary	1.63%
Consumer staples	1.35%
Communication services	0.72%
Information technology	0.52%

Price to NAV Ratio



Source: <https://app.koyfin.com/charts/g/et-yh6fpp?i=gf.dy>

Dividend Yield & MACD

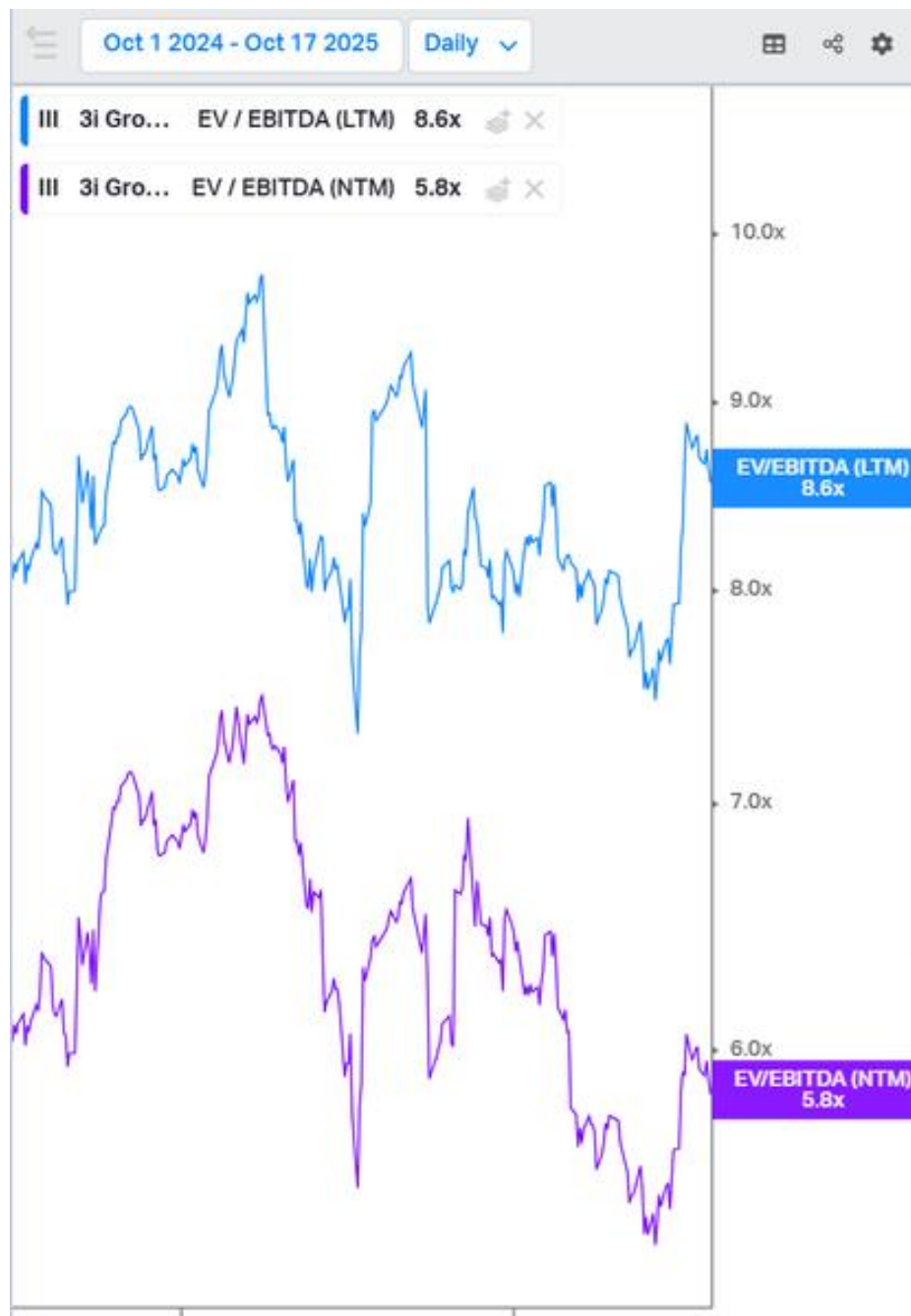
- Momentum: PSP is currently experiencing strong bearish (downward) momentum.
- Income: The dividend yield has recently become very volatile and has settled at a new, lower level of 5.45% compared to its range for most of the past year.



Source: <https://app.koyfin.com/charts/g/et-yh6fpp?i=gf.dy>

EV/EBITDA Multiples of Top 2 Firms

3i Group Plc



EQT Corporation

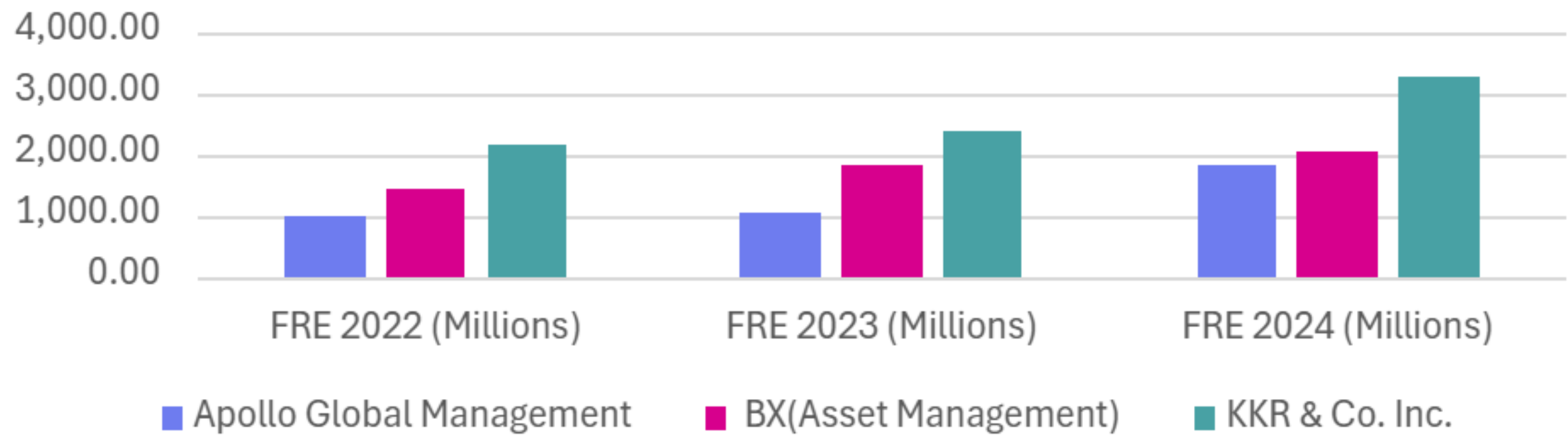


Source: <https://app.koyfin.com/charts/g/eq-9c4tik>

Fee Related Earnings (FRE) Impact

Top 3 Holding Private Equity Companies' Annual Report

Company	FRE 2022 (Millions)	FRE 2023 (Millions)	FRE 2024 (Millions)
Apollo Global Management	1,008.70	1,058.70	1,828.20
BX(Asset Management)	1,410.00	1,788.00	2,063.00
KKR & Co. Inc.	2,167.40	2,383.80	3,267.80



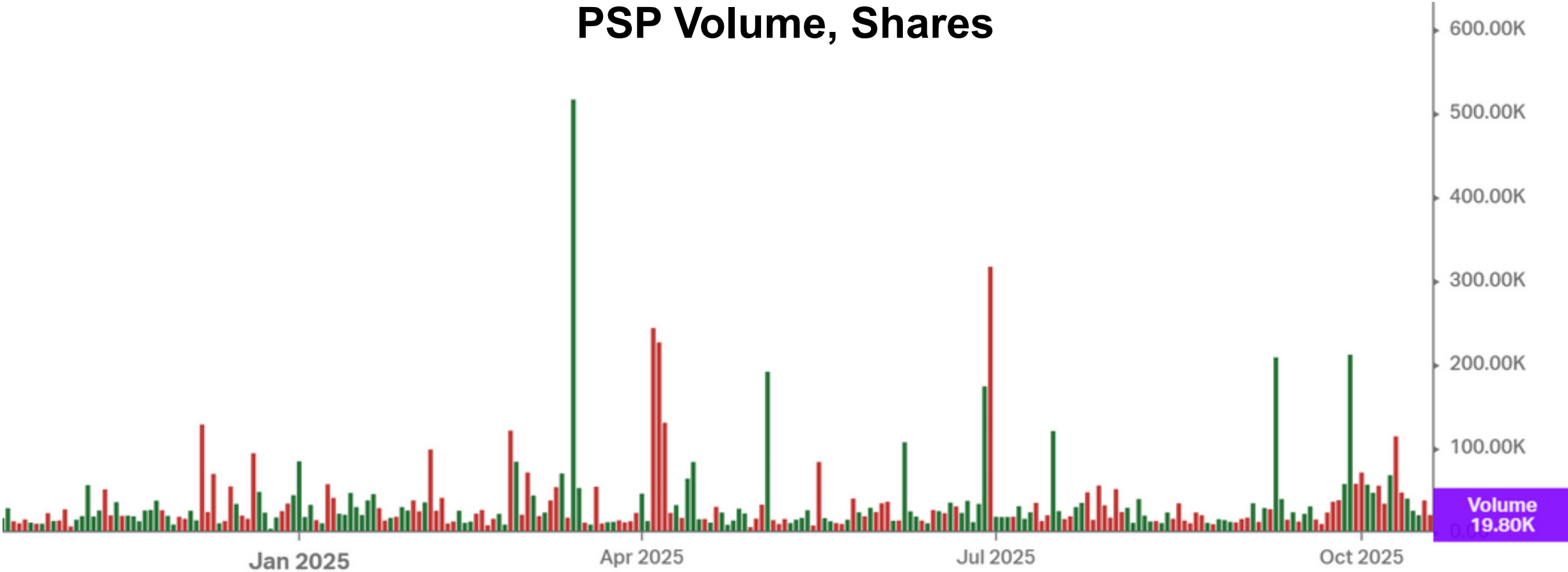
Source:

<https://www.sec.gov/ix?doc=/Archives/edgar/data/0001393818/000119312525042469/d912273d10k.htm>

<https://www.sec.gov/ix?doc=/Archives/edgar/data/0001858681/000185868125000034/apo-20241231.htm>

<https://www.sec.gov/ix?doc=/Archives/edgar/data/0001404912/000140491225000015/kkr-20241231.htm>

Trading Liquidity



Source: <https://app.koyfin.com/charts/g/et-yh6fpp>

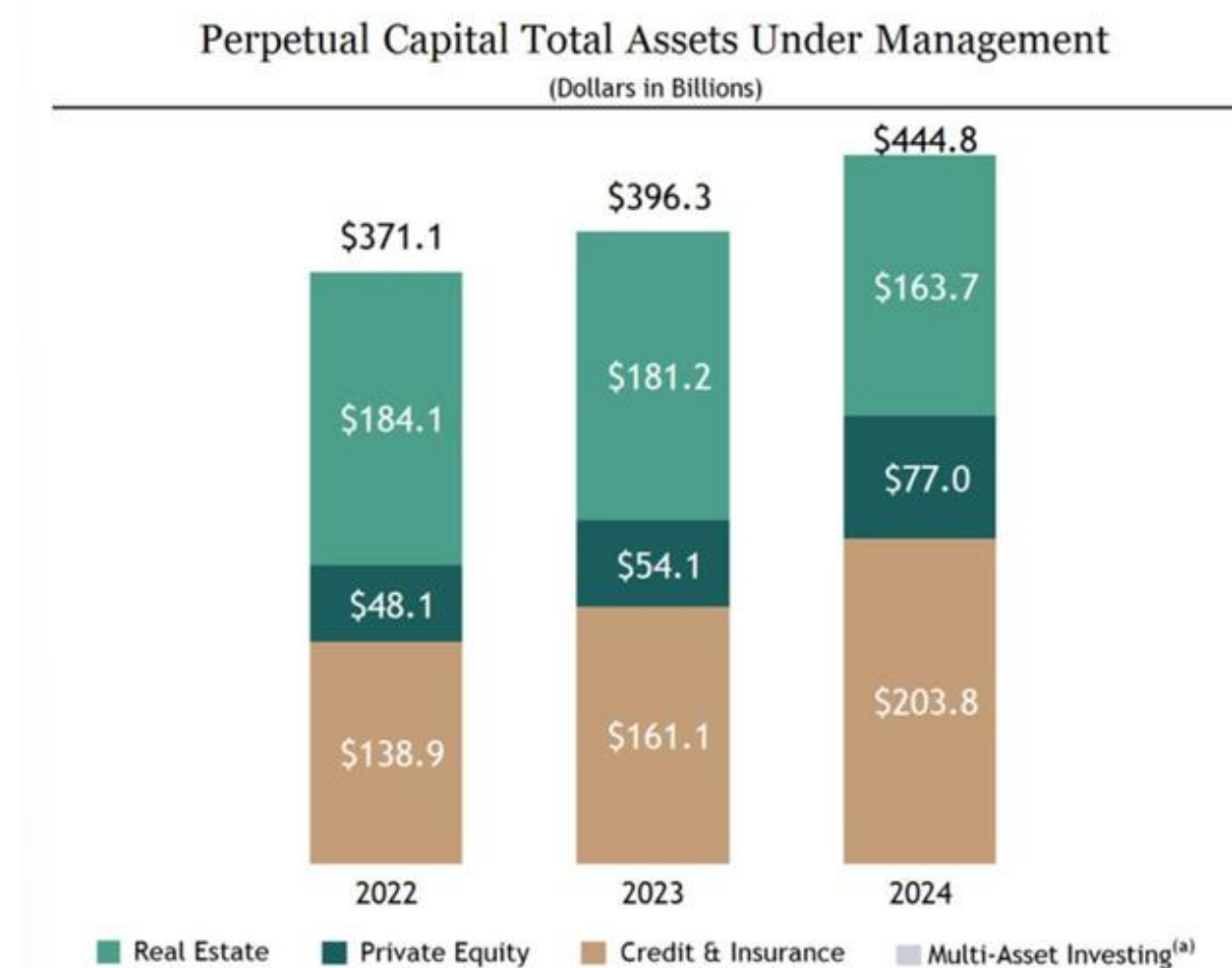
Short Interest

This chart illustrates a dynamic and fluctuating level of bearish sentiment on PSP over the past year. While there were periods of intense pessimism, the current short interest level has subsided to a more moderate 1.30%.



Source: <https://app.koyfin.com/charts/g/et-yh6fpp>

Real Estate & Credit Exposure Risks



- PSP faces significant real estate and credit risks stemming from its exposure to PE firms invested in property and leveraged corporate assets.
- Persistent high interest rates lower real estate valuations and increase debt refinancing costs.
- Tight credit markets, rising loan defaults, and impairments across portfolio companies amplify downside risk to NAV and share price.

Source: <https://www.ropesgray.com/en/insights/alerts/2025/03/us-pe-market-recap>

<https://www.sec.gov/ix?doc=/Archives/edgar/data/0001393818/000119312525042469/d912273d10k.htm>

Real Estate & Credit Exposure Risks

U.S. Middle Market Lending

- Loan Volume: U.S. middle market loan volume increased 49% in 2024, reaching \$297 billion.
- LBO Lending: U.S. mid-market LBO lending totaled \$55 billion in 2024, marking a 66% year-over-year increase. Direct lending accounted for 90% of LBO lending, up from 36% in 2014.
- Equity Contributions: Equity contributions on middle market LBO deals declined to 55% in Q4 2024, below the five-year average of 59%.

Source: <https://www.ropesgray.com/en/insights/alerts/2025/03/us-pe-market-recap>

<https://www.sec.gov/ix?doc=/Archives/edgar/data/0001393818/000119312525042469/d912273d10k.htm>

Regulatory Risk Assumptions

Regulatory Risk	Description	Probability	Impact	Weighted Risk Score	PSP Implications
SEC Fund Disclosure & Oversight	Increased transparency & liquidity reporting requirements	4	4	16	Higher compliance costs reduce fee growth & earnings
Global Tax & Reporting Standards	OECD/EU tax harmonization on carried interest & offshore reporting	3	4	12	Reduces after-fee profits, complicates global structures
Governance & ESG Accountability	Stricter ESG & governance rules add operational risk	4	3	12	Raises cost base, reputational risk impacts valuations
Systemic Risk Monitoring	Large PE firms flagged as systemic, may face capital constraints	3	4	12	Limits leverage and liquidity; reduces return on equity
Market Conduct & Antitrust	DOJ/FTC probes slow deal approvals	4	3	12	Reduced deal flow impacts carried interest & fee revenue
Cross-Border Regulation	Divergent US/EU fund marketing & privacy rules	3	3	9	Legal & compliance costs rise
Leverage & Margin Controls	Tighter oversight of derivatives & leverage limits	4	4	16	Limits investment flexibility, may increase volatility

Macro Risk Assumptions

Macro Risk	Description	Probability	Impact	Weighted Risk Score	PSP Implications
Interest Rate Environment	Sustained high real rates (4–5%) increase discount rates	5	5	25	Compresses buyout valuations, pressures NAV and returns
Credit Market Health	Leveraged loan issuance down 25%, spreads wider by ~100bps	4	4	16	Rising debt costs compress portfolio company earnings
Fundraising & Liquidity	Slower capital recycling; growth of continuation funds	4	4	16	Restricts inflows and fee growth for PE managers
Inflation & Trade Policy	Persistent inflation & tariff volatility raise cost pressures	4	3	12	Margin pressure on portfolio companies
Global Growth Outlook	Slowing US/China growth; mild EU recovery	3	3	9	Dampen deal flow, valuations, and earnings
Political & Policy Uncertainty	Geopolitical & tariff uncertainty impact sentiment	4	3	12	Drives episodic risk-off volatility
Currency & Rate Volatility	Strong USD and FX volatility create translation risk	3	3	9	NAV and earnings translation fluctuations
Sector Rotation	Capital shifting from buyouts to credit, infrastructure	4	4	16	Lower inflows for PSP holdings' strategies

Performance Metrics of PSP ETF

Variable	Abbreviation	PSP	MSCI ACWI	PSP Implications
Risk Free rate	Rf(US 10Y Treasury)	4.01%	4.01%	Common benchmark
Annual Return	Rp	16.22%	20.29%	PSP underperforms its benchmark
Annual Standard Deviation	σ	22.68%	12.35%	PSP is significantly more volatile
Downside Deviation	σ_D	15.46%	9.14%	PSP has greater downside risk
Sharpe Ratio(3Y)		0.54	1.32	Investors are not adequately compensated for higher volatility
Sortino Ratio(3Y)		0.79	1.78	Weak downside-adjusted performance

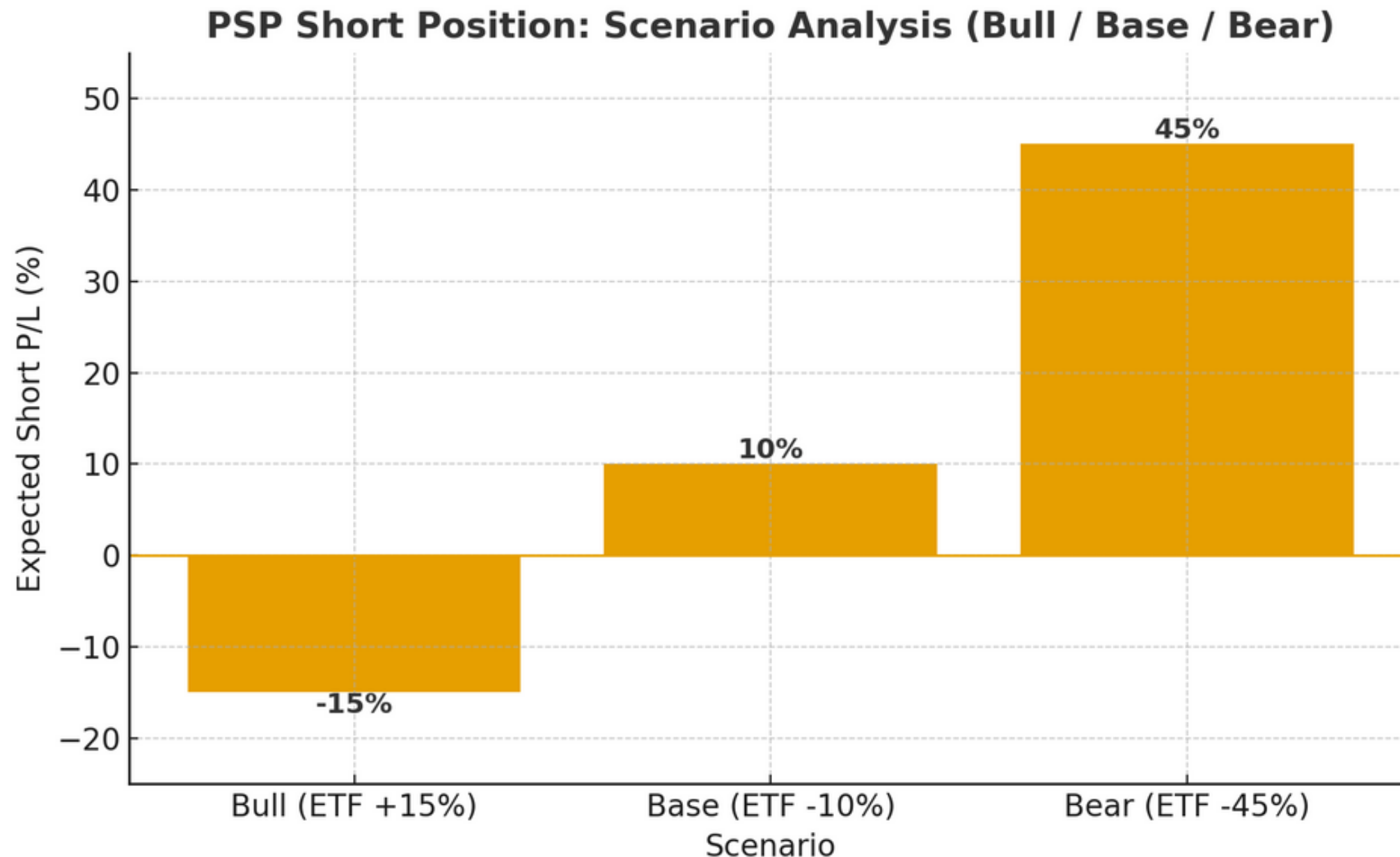
Source: <https://www.investing.com/etfs/powershares-listed-private-equity-historical-data>;
<https://www.investing.com/indices/msci-world-stock-historical-data>; <https://www.cnbc.com/quotes/US10Y>

Comparative Analysis with Peer ETFs

Metric	PSP (Invesco Global Listed PE ETF)	PEX (ProShares Global Listed PE ETF)	IIM (BlackRock PE ETF)	PIO (IPATH PE Index)
Expense Ratio (%)	1.44%	3.13%	1.25%	0.85%
Price-to-NAV Premium	~1.1x	~1.0x	~1.05x	~0.95x
Dividend Yield (%)	~4.6%	~11.6%	~3.2%	~2.8%
Total Net Assets (USD M)	\$326M	\$26M	\$120M	\$190M
Average Daily Trading Volume	~150,000 shares	~40,000 shares	~70,000 shares	~60,000 shares
Top 10 Holdings Concentration (%)	~46%	~55%	~40%	~60%
1-Year Price Return	~3.25% (flat/low growth)	~-7.4% (negative)	~-3.6%	~6.1%
Historical Volatility (1-Year %)	~22%	~18%	~20%	~16%

Source: <https://seekingalpha.com/symbol/PSP/peers/comparison>;
<https://etfdb.com/etf/PSP/>

Scenario Analysis



This chart shows the expected downside under 3 conditions
PSP could underperform as credit conditions tighten and private-equity deal activity weakens.

Source: https://www.dividend.com/etfs/psp-invesco-global-listed-private-equity-etf/?utm_source

Price Projections

Scenario	Macro Drivers	Historical Assumption	PSP Move	Projected Price
Bull (+15%)	Rapid rate cuts, spread compression, strong deal rebound	2023 rebound (+37%)	15%	$67 \times (1 + 0.15) = 77$
Base (-10%)	Flat rates/spreads, moderate deals	Mean-reversion pattern	-10%	$67 \times (1 - 0.10) = 60$
Bear (-45%)	Recession, spread blowout, PE freeze	2022 crash (-39%) & 2008 analog	-45%	$67 \times (1 - 0.45) = 37$

Source:

https://www.financecharts.com/etfs/PSP/performance?utm_source

Investment Execution Plan

Component	Key Actions/Parameters
Position Sizing	Initiate at 2–5% portfolio with scaling
Entry Criteria	Enter when Price-to-NAV premium >1.1x
Entry Strategy	Use 3 tranche limit orders to average price
Stop Loss	Set stop losses at 5–8% above entry
Hedging	Hedge with private credit ETFs or options
Monitoring	Weekly review of valuation, fundraising, macro signals
Exit Strategy	Exit partially or fully when premium compresses <1.0x or earnings improve
Execution Notes	Confirm borrow availability, manage margin and dividend risk

Financial Instrument

- The short will be executed through the direct sale of PSP shares via brokerage margin. This involves borrowing shares and selling them in the market with the obligation to repurchase later.
- PSP's average daily volume (~150,000 shares) supports efficient execution with manageable slippage. Borrow availability and borrow cost, around 2% annualized, will be closely monitored for cost control.
- Entry will be staged using limit orders to minimize price impact. This approach provides direct exposure to PSP's price declines and allows precise position management.

Investment Summary

- This project analyzes and recommends a short position in the Invesco Global Listed Private Equity ETF (PSP), which tracks the Red Rocks Global Listed Private Equity Index.
- The ETF invests primarily in publicly traded private-equity managers such as KKR, Blackstone, Apollo, EQT, and Partners Group, giving investors indirect exposure to global buyouts, venture capital, and private credit.
- Although private-equity deal value and fundraising showed a modest rebound in 2025, fundamentals remain weak. Global fundraising has dropped more than 40% from the 2021 peak, while EBITDA purchase multiples remain elevated at 12–15×, indicating valuation risk.
- Rising financing costs, slower exits, and aging portfolios continue to pressure returns. PSP's 93.9% exposure to financials and high concentration in its top 10 holdings (~46%) amplify downside risk.
- With slowing liquidity, higher leverage, and fragile credit conditions, PSP appears overvalued, supporting a short thesis as valuations normalize and credit spreads widen.